CHAPTER 3

METHODOLOGY

This chapter presents the design and methodology for developing the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing. It includes the selected research approach, the most appropriate Software Development Life Cycle (SDLC) model, and the system requirements specifications.

3.1 Systems Development Method

The systems development method refers to the framework used to structure, plan, and control the process of developing the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing. The proponents used the Agile methodology, specifically an incremental development model, which allows flexibility and continuous feedback throughout the system development process.

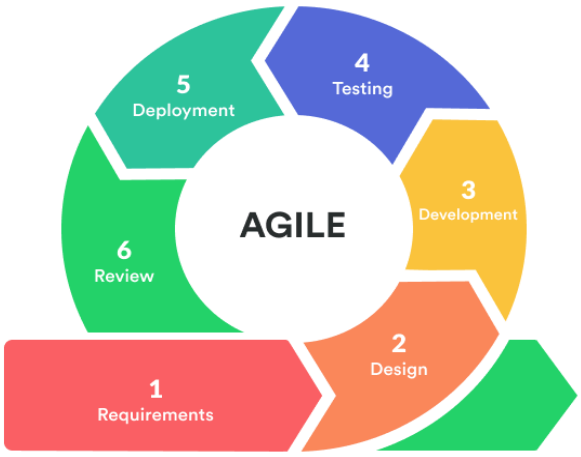


Figure 1: Agile development method of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing

Plan

In the Plan phase, the researchers collaborated with the management, cashier, and inventory staff of Meryl Shoes to define the project's scope, goals, and necessary features. The team began by conducting interviews and gathering data about the current system, including manual recording using logbooks and receipts, separate inventory monitoring, and reporting challenges. This process helped identify issues such as discrepancies in sales records, a lack of real-time monitoring, and inefficiencies in inventory tracking. Additionally, the researchers defined the project's objectives and system requirements, ensuring that the proposed integrated POS-driven inventory system with data analytics would address the business needs.

Design

During the Design phase, the gathered requirements were translated into system architecture, database design, and user interface wireframes. The researchers designed the system structure to support key modules such as POS transactions, inventory management, reporting, and analytics. The database schema was created to handle sales, product, inventory, and user data efficiently. The user interface was designed to be simple and user-friendly, considering the roles of the cashier, inventory staff, and manager. A design review was conducted to ensure that the system meets all functional and operational requirements identified during the planning phase.

Develop

In the Develop phase, the system was developed using an incremental approach through short iterations. Each iteration focused on implementing core features such as sales processing, automatic stock deduction, inventory monitoring, and report generation. The development team set up the system environment, implemented the database, and developed both front-end and back-end components. The POS module was created for transaction processing, while the inventory module handled stock updates and monitoring. Additional features, such as analytics for best-selling and slow-moving products, were also developed to support decision-making.

Test

The Test phase was conducted alongside development to ensure system quality and functionality. Unit testing was performed on individual components, followed by integration testing to verify that all modules work together properly. User acceptance testing was conducted with the cashier, inventory staff, and manager to evaluate usability and performance. The system was assessed based on functionality, accuracy, usability, and efficiency. Identified errors and issues were fixed, and improvements were made to ensure the system meets the required standards.

Deploy

After successful testing, the system was deployed in a live environment. The system was first introduced in a test setup where users evaluated its functionality and usability. Once validated, it was implemented for actual use in business operations. Training was provided to the cashier, inventory staff, and manager to ensure proper usage of the system. A deployment review was also conducted to confirm that the system operates correctly and meets the expected requirements.

Review

In the Review phase, feedback was collected from users to evaluate the effectiveness of the system in improving transaction recording, inventory management, and reporting. The researchers assessed how well the system addressed the identified problems, including manual processes and lack of integration. Any remaining issues were documented, and recommendations for system improvement were identified. System documentation was also prepared for future reference and maintenance.

Launch

In the Launch phase, the final version of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing was officially implemented. The system was designed to support real-time transaction processing, accurate inventory monitoring, and data-driven

decision-making. The system was fully deployed for business use, and continuous support was provided to ensure smooth operation and user assistance.

3.2 Operational Framework

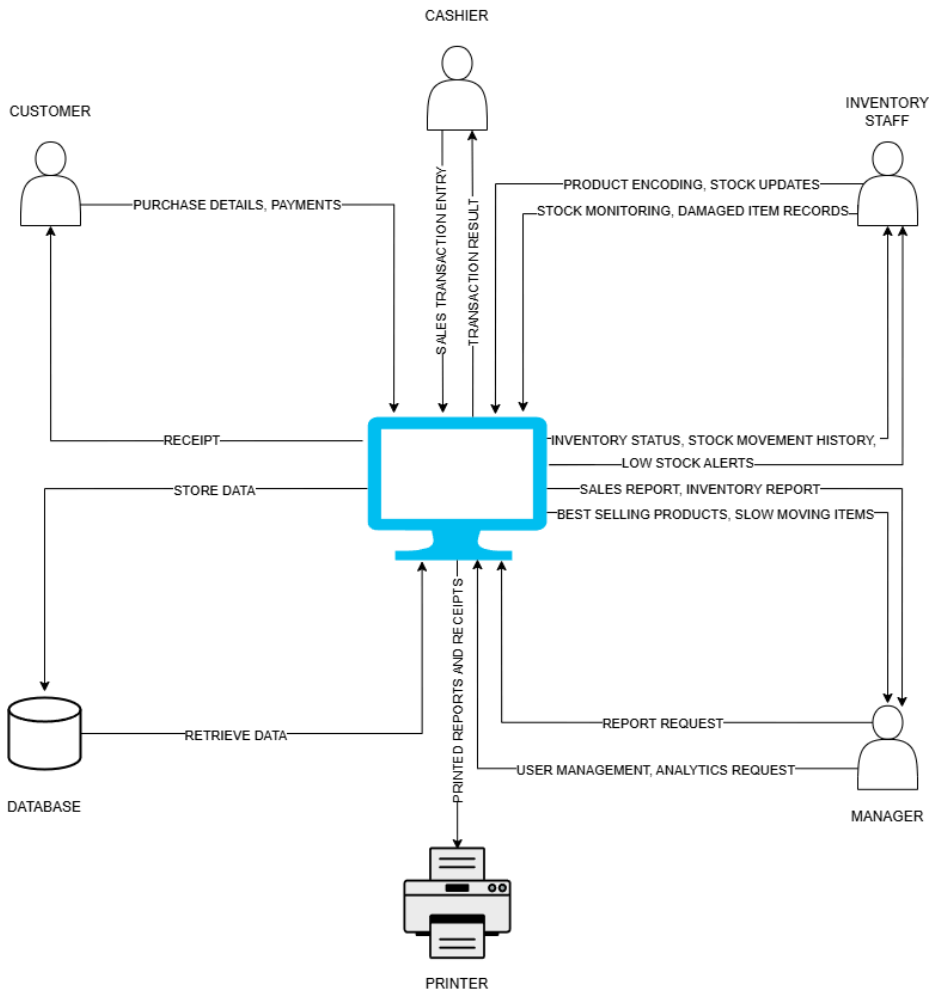


Figure 2 shows the operational framework of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing. It illustrates how the customer, cashier, inventory staff, and manager interact with the system to input, process, and access business information.

The customer provides purchase details and payment, and receives a receipt after the transaction is completed. The cashier handles sales transaction entry and receives the transaction result from the system. The inventory staff manages product encoding, stock updates, stock monitoring, and damaged item records, while also receiving inventory status, stock movement history, and low stock alerts from the system. The manager sends report requests, user management tasks, and analytics requests, and receives outputs such as sales reports, inventory reports, best-selling products, and slow-moving items.

All information is stored in a central database, making sales and inventory data easier to manage, retrieve, and update. The printer is used to produce hard copies of receipts and reports. Through this framework, the system improves transaction processing, inventory monitoring, reporting, and analytics, making retail operations more efficient, accurate, and responsive to business needs.

3.3 Context Diagram

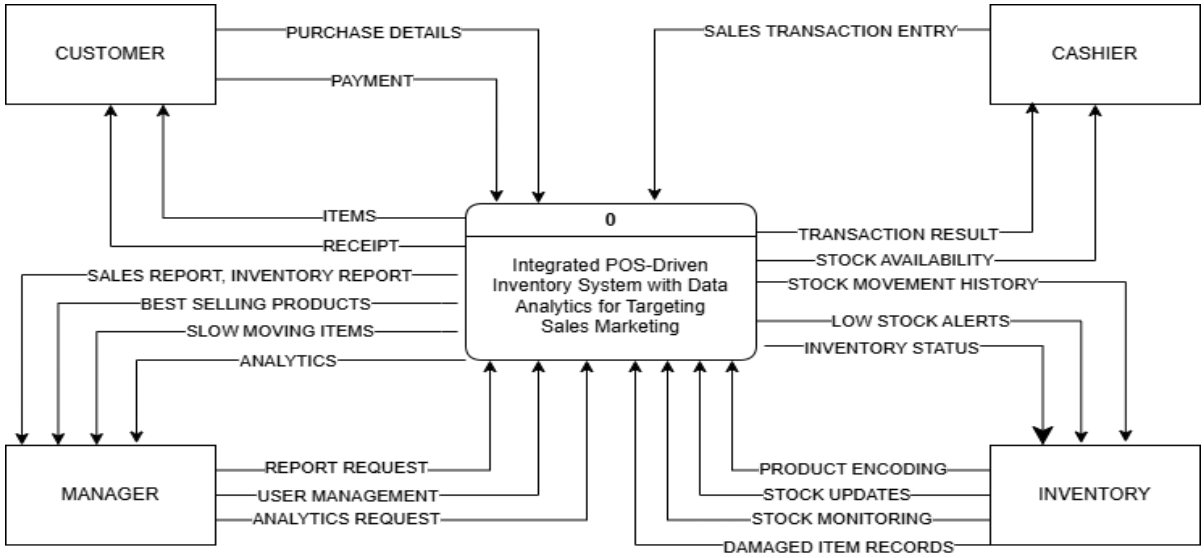
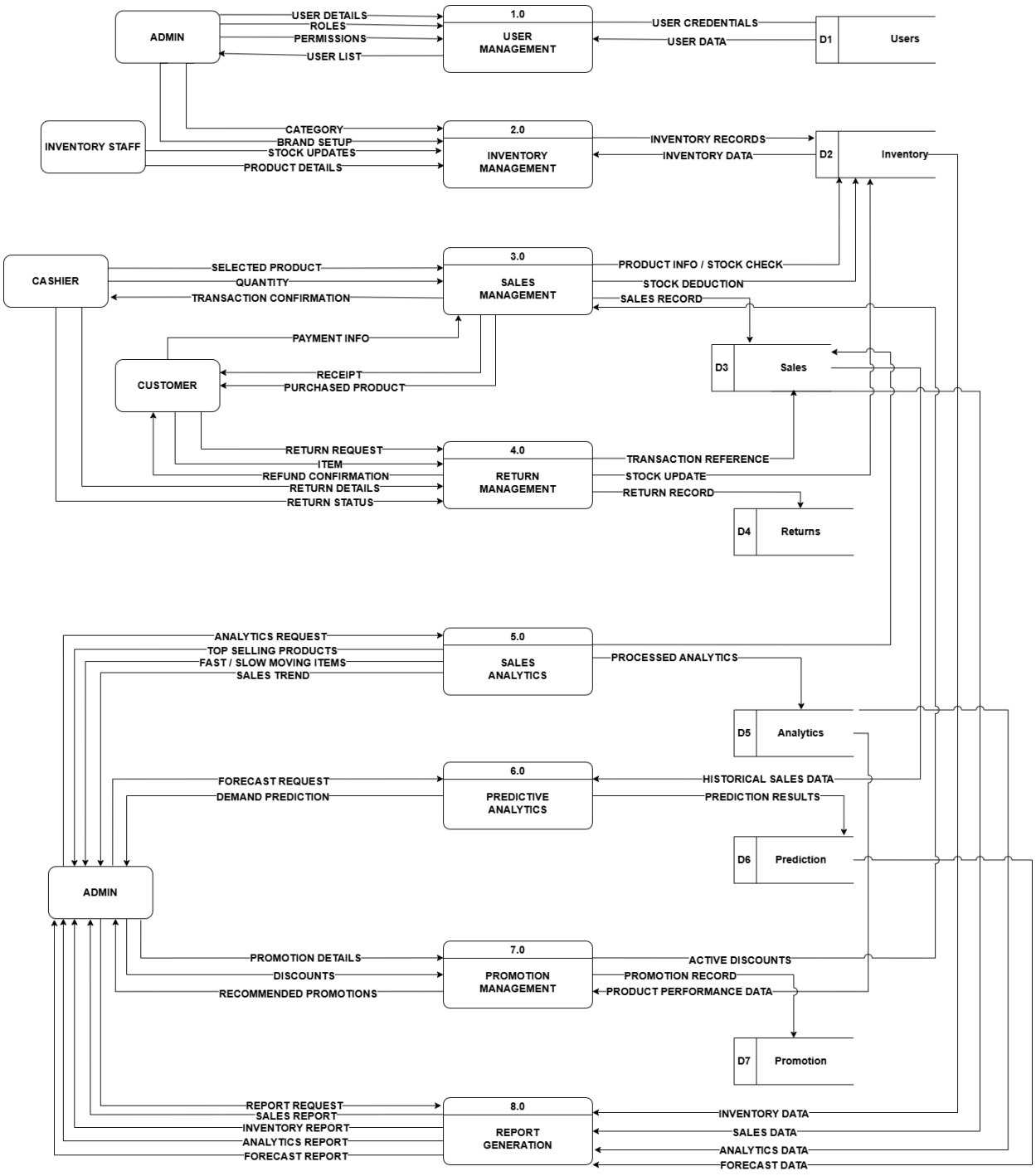


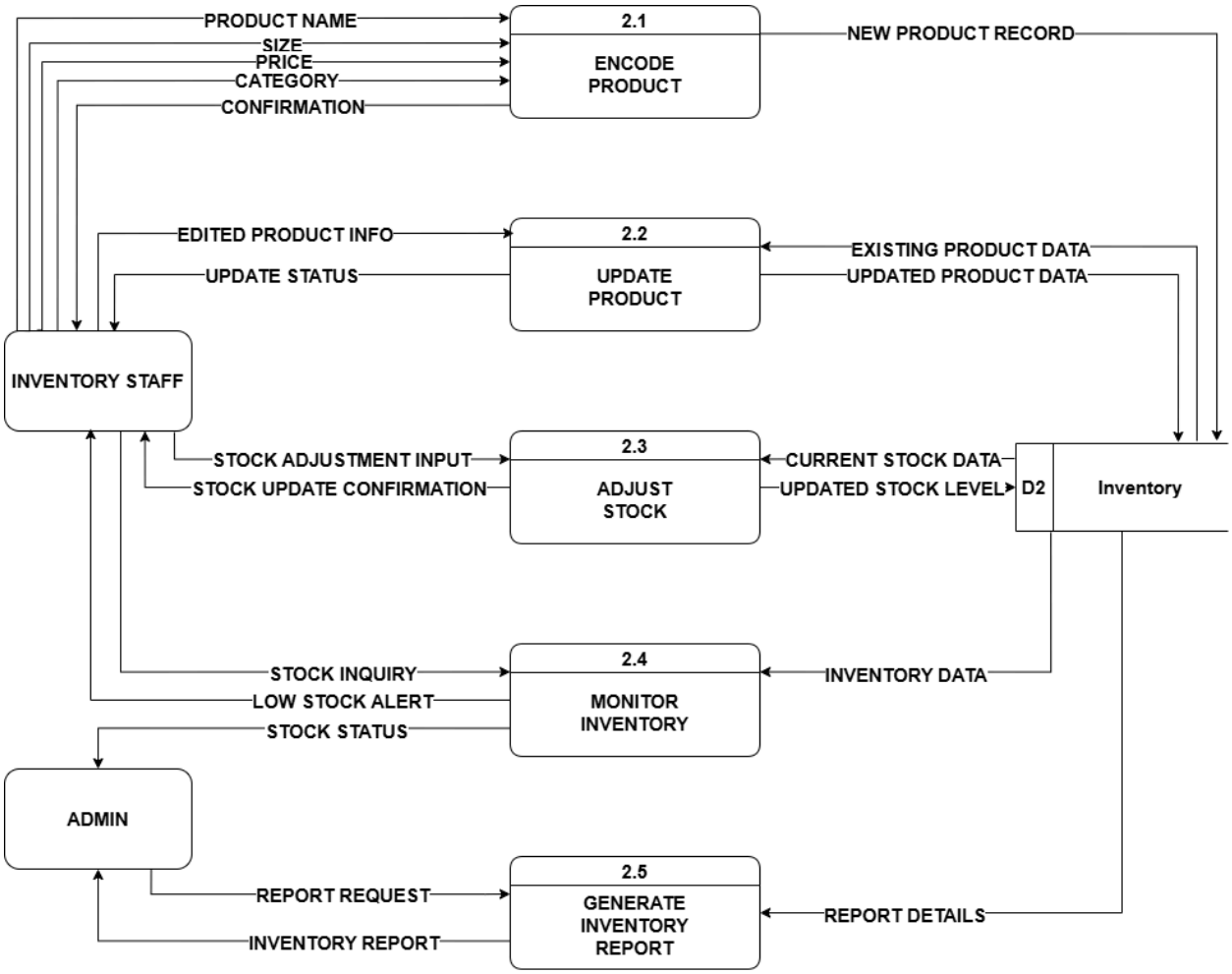
Figure 3: Context Diagram of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing

Figure 3 shows the context diagram of the developed Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing, illustrating how data from the customer, cashier, inventory staff, and manager is collected, processed, and transformed into sales transactions, inventory updates, and analytical reports to support business operations and decision-making.

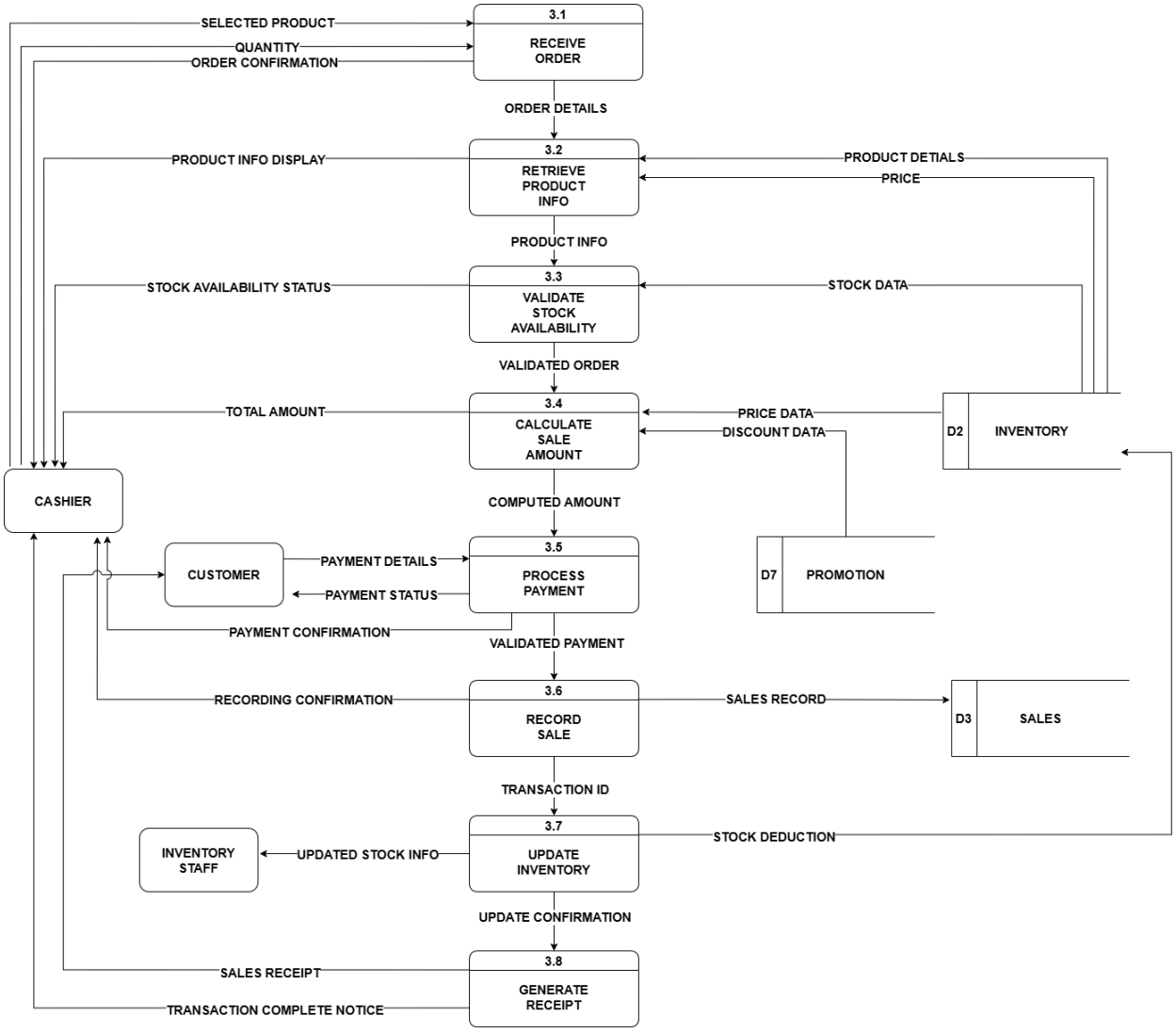
3.4 Data Flow Diagram



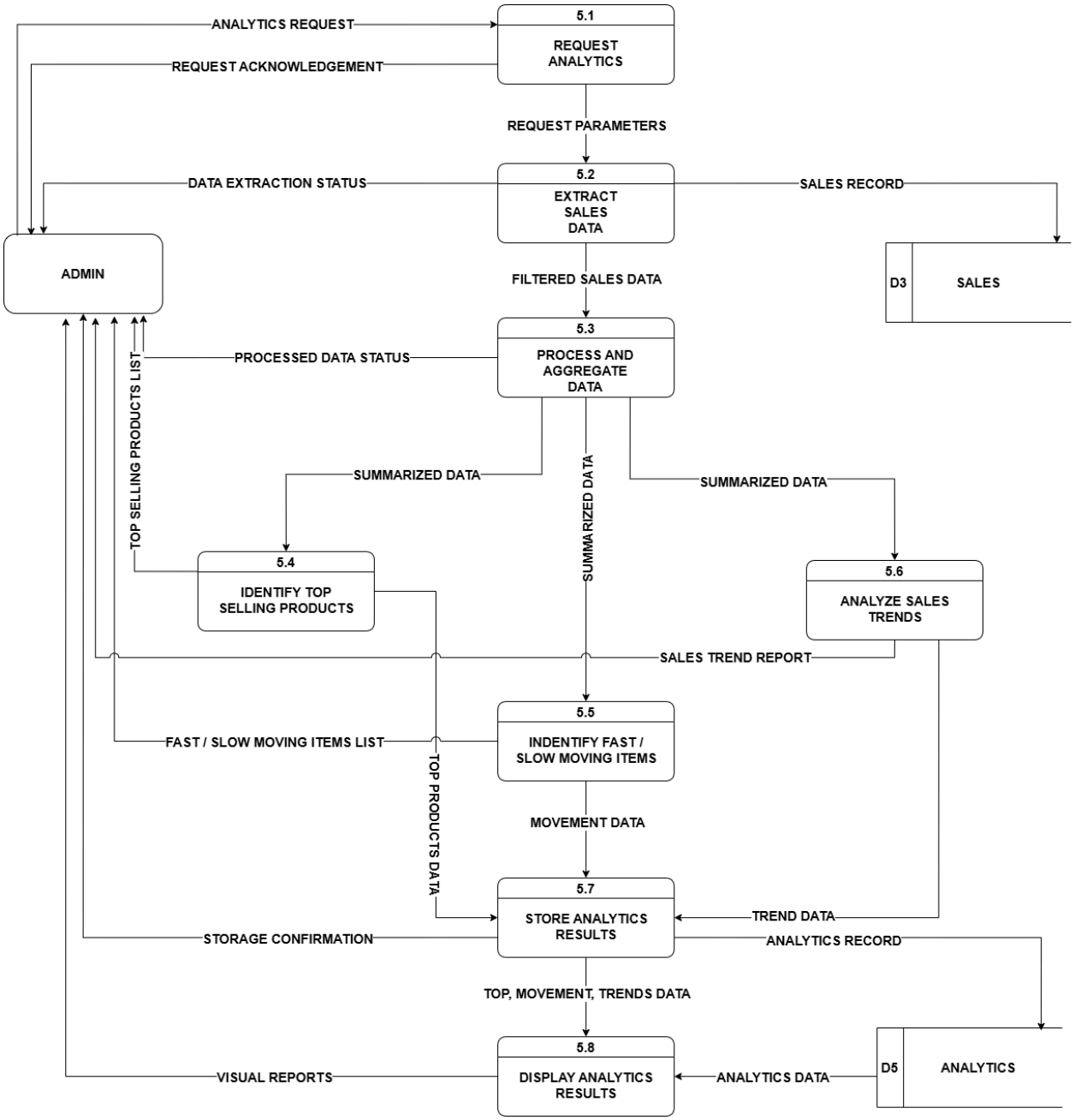
LVL 1



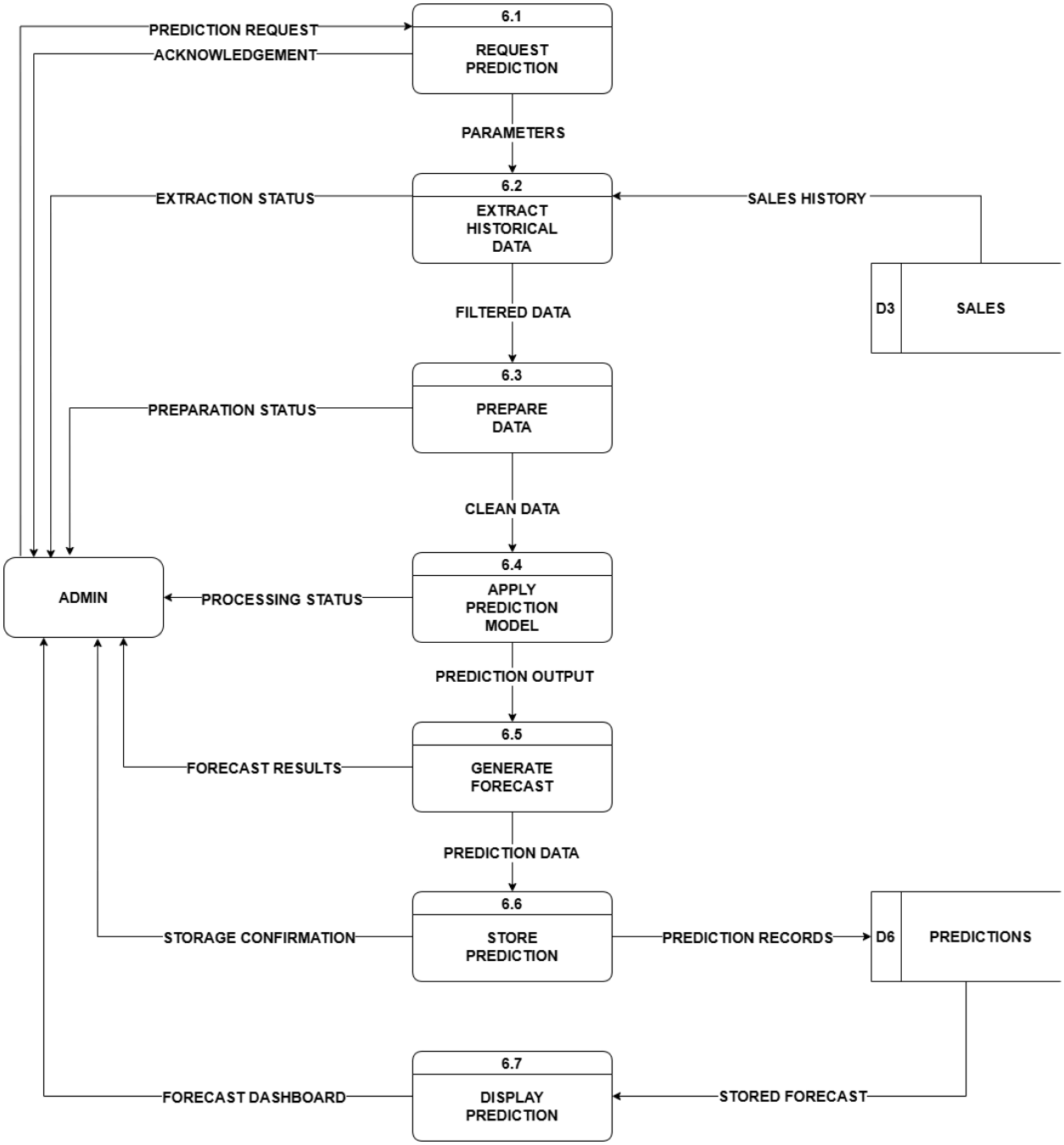
LVL 2 INVENTORY


3.1
RECEIVE ORDER
SELECTED PRODUCT
QUANTITY
ORDER CONFIRMATION
ORDER DETAILS
3.2
RETRIEVE PRODUCT INFO
PRODUCT INFO DISPLAY
PRODUCT DETIALS
PRICE
PRODUCT INFO
3.3
VALIDATE STOCK AVAILABILITY
STOCK AVAILABILITY STATUS
STOCK DATA
VALIDATED ORDER
3.4
CALCULATE SALE AMOUNT
TOTAL AMOUNT
PRICE DATA
DISCOUNT DATA
D2
INVENTORY
CASHIER
COMPUTED AMOUNT
3.5
PROCESS PAYMENT
CUSTOMER
PAYMENT DETAILS
PAYMENT STATUS
D7
PROMOTION
PAYMENT CONFIRMATION
VALIDATED PAYMENT
3.6
RECORD SALE
RECORDING CONFIRMATION
SALES RECORD
D3
SALES
TRANSACTION ID
3.7
UPDATE INVENTORY
INVENTORY STAFF
UPDATED STOCK INFO
STOCK DEDUCTION
UPDATE CONFIRMATION
3.8
GENERATE RECEIPT
SALES RECEIPT
TRANSACTION COMPLETE NOTICE

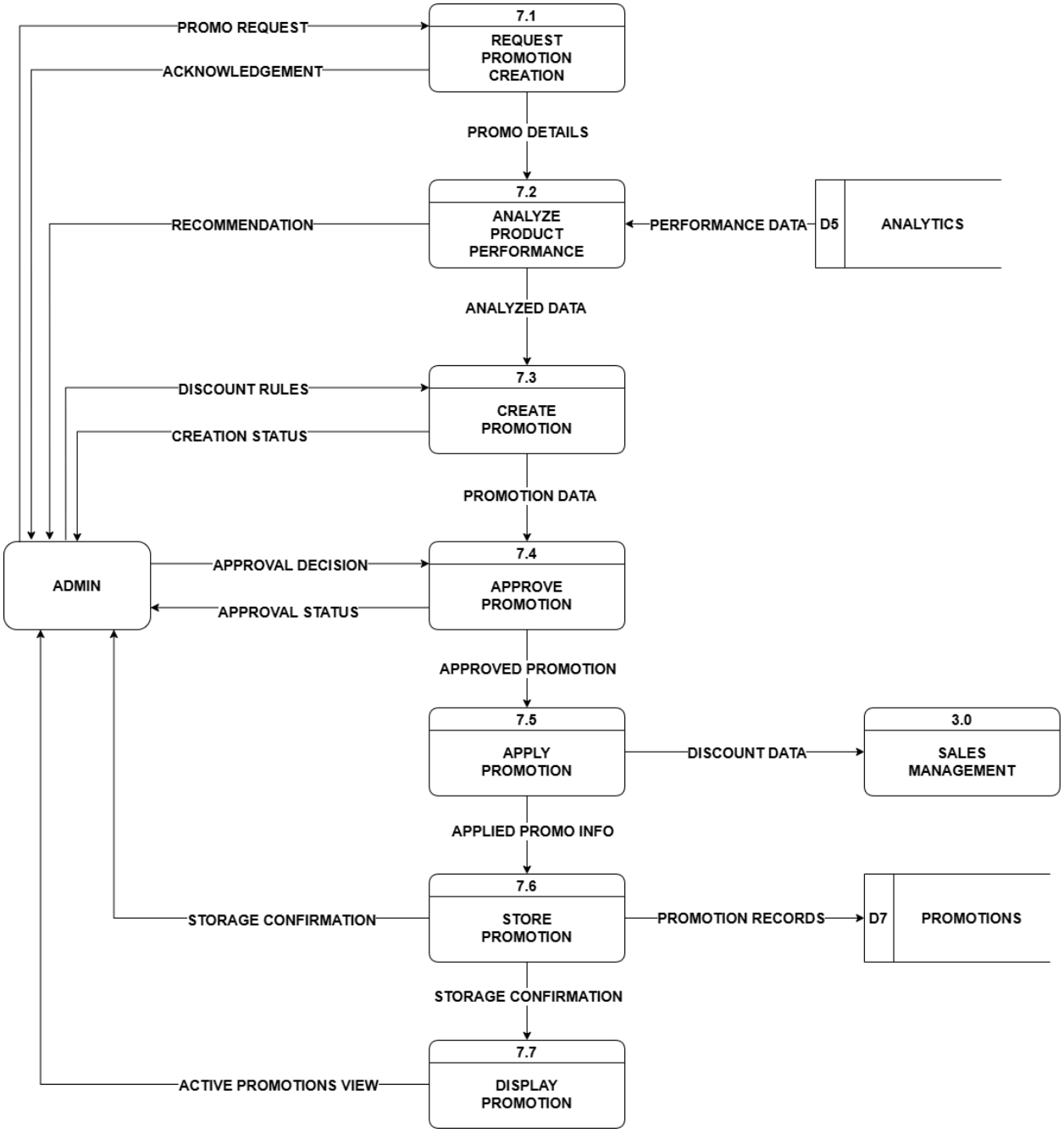
LVL 2 SALES



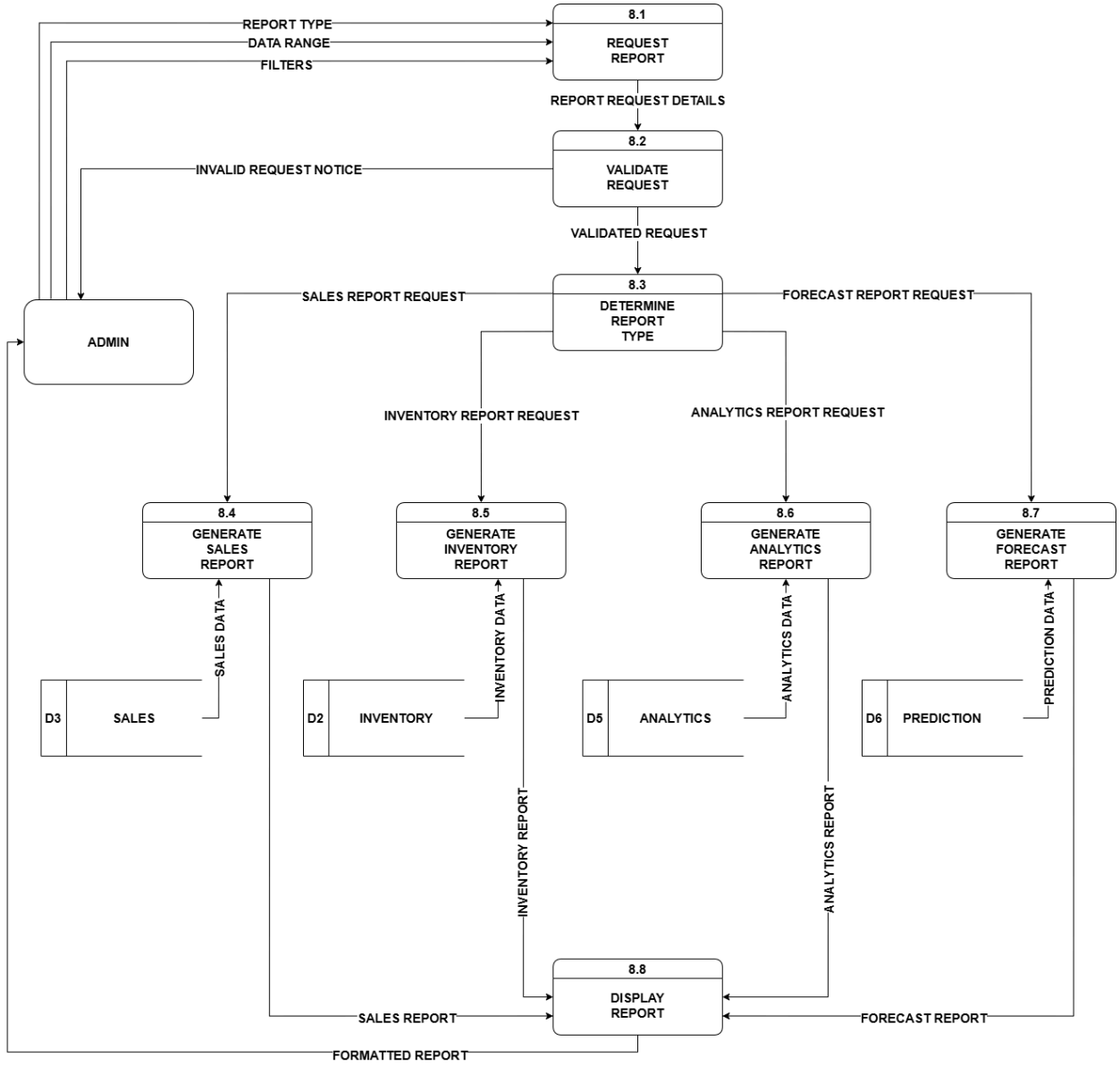
LVL 2 ANALYTICS



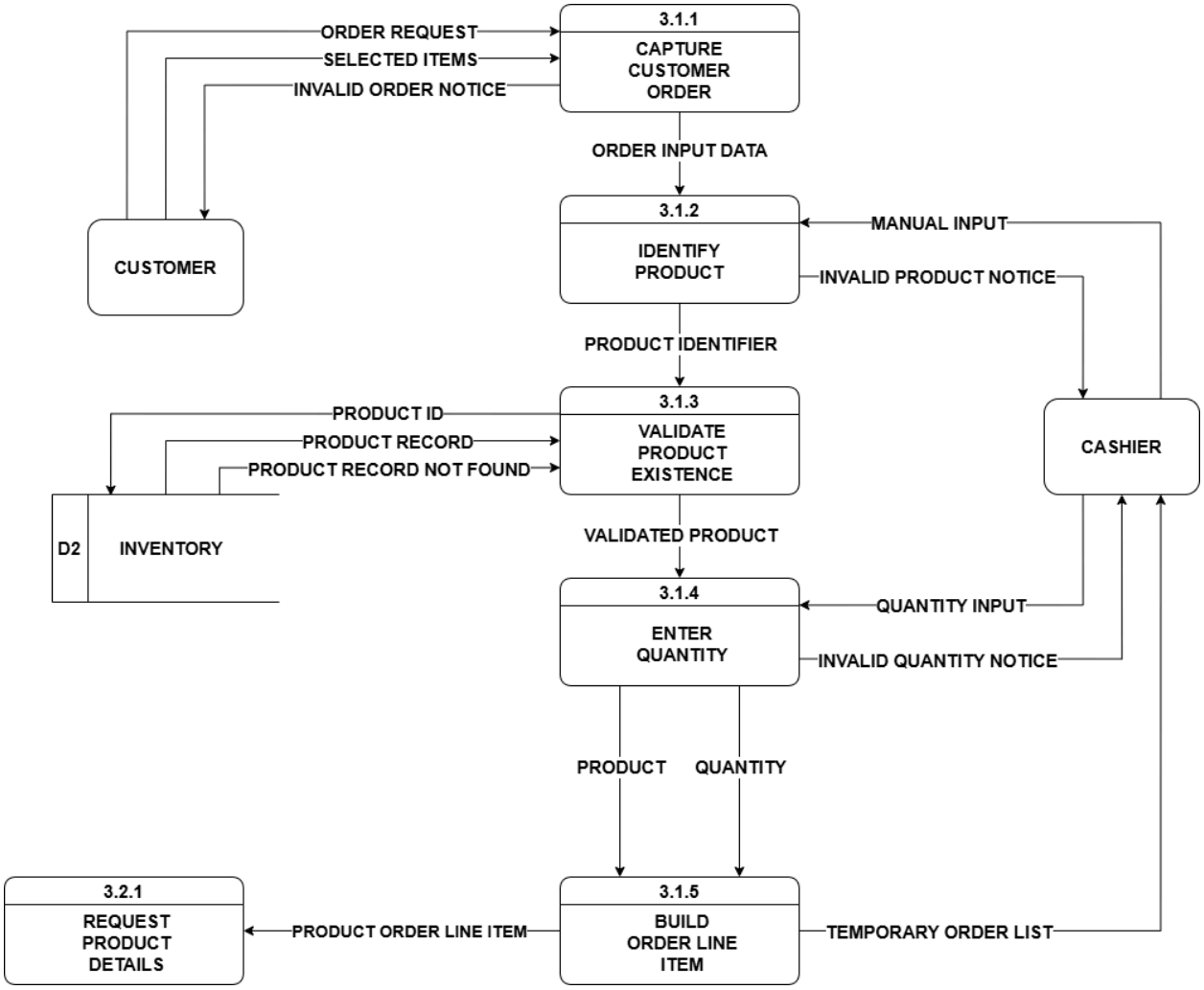
LVL 2 PREDICTION



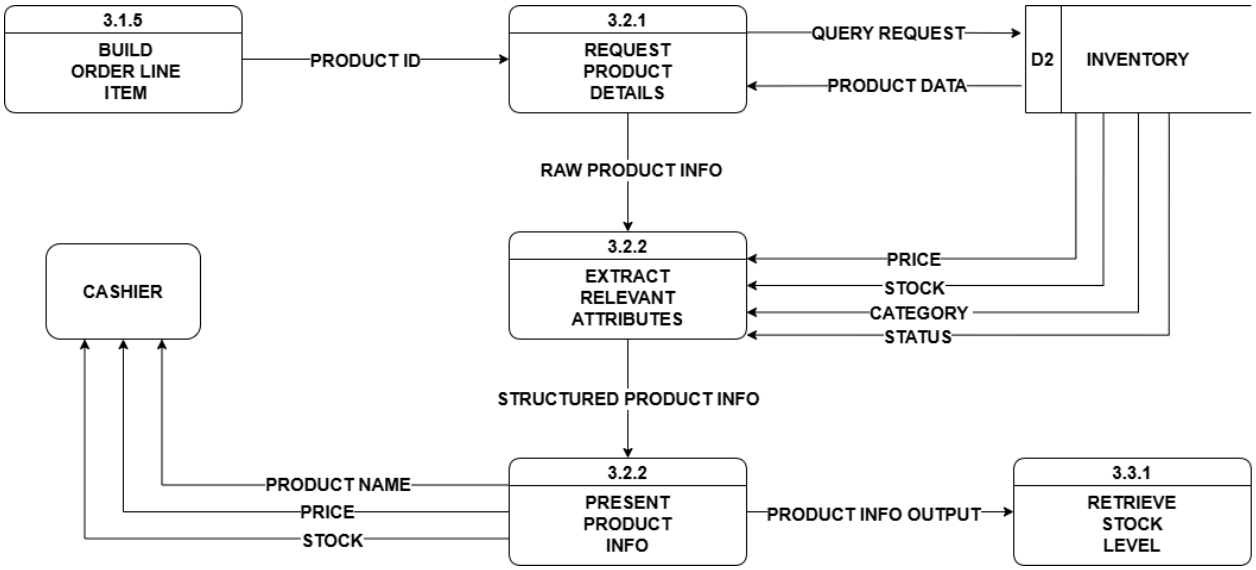
LVL 2 PROMOTION


8.1
REQUEST REPORT
REPORT TYPE
DATA RANGE
FILTERS
REPORT REQUEST DETAILS
8.2
VALIDATE REQUEST
INVALID REQUEST NOTICE
VALIDATED REQUEST
8.3
DETERMINE REPORT TYPE
SALES REPORT REQUEST
FORECAST REPORT REQUEST
ADMIN
INVENTORY REPORT REQUEST
ANALYTICS REPORT REQUEST
8.4
GENERATE SALES REPORT
8.5
GENERATE INVENTORY REPORT
8.6
GENERATE ANALYTICS REPORT
8.7
GENERATE FORECAST REPORT
SALES DATA
INVENTORY DATA
ANALYTICS DATA
PREDICTION DATA
D3
SALES
D2
INVENTORY
D5
ANALYTICS
D6
PREDICTION
INVENTORY REPORT
ANALYTICS REPORT
8.8
DISPLAY REPORT
SALES REPORT
FORECAST REPORT
FORMATTED REPORT

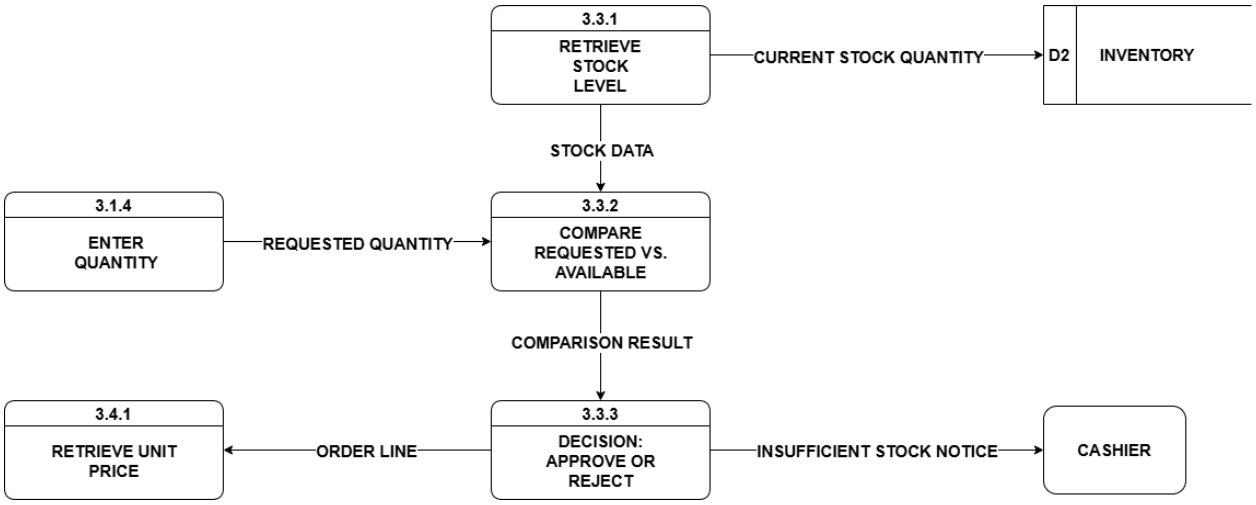
LVL 2 REPORT



LVL 3 (3.1 RECEIVE ORDER)



LVL 3 (3.2RETRIEVE PRODUCT INFORMATION)



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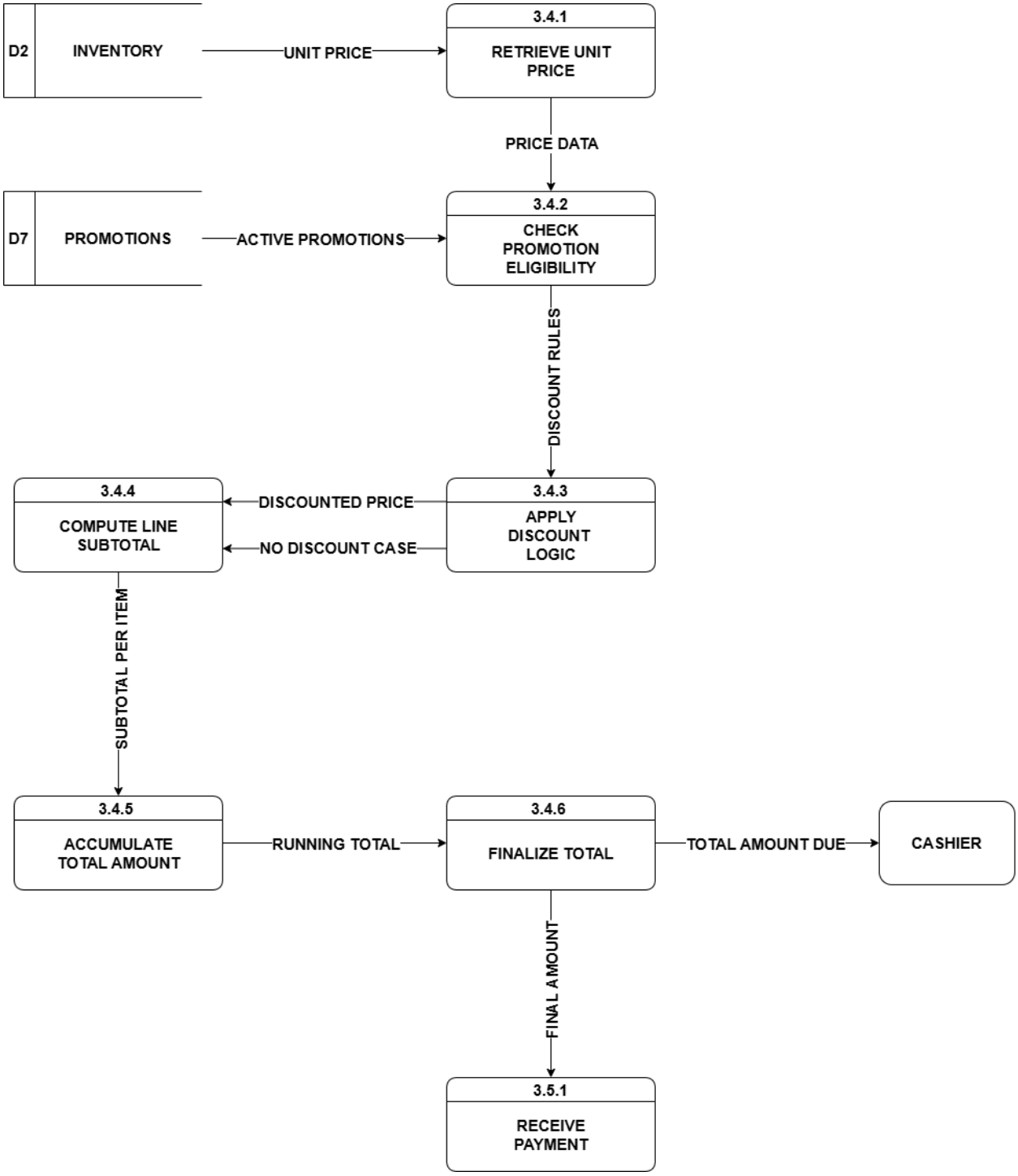
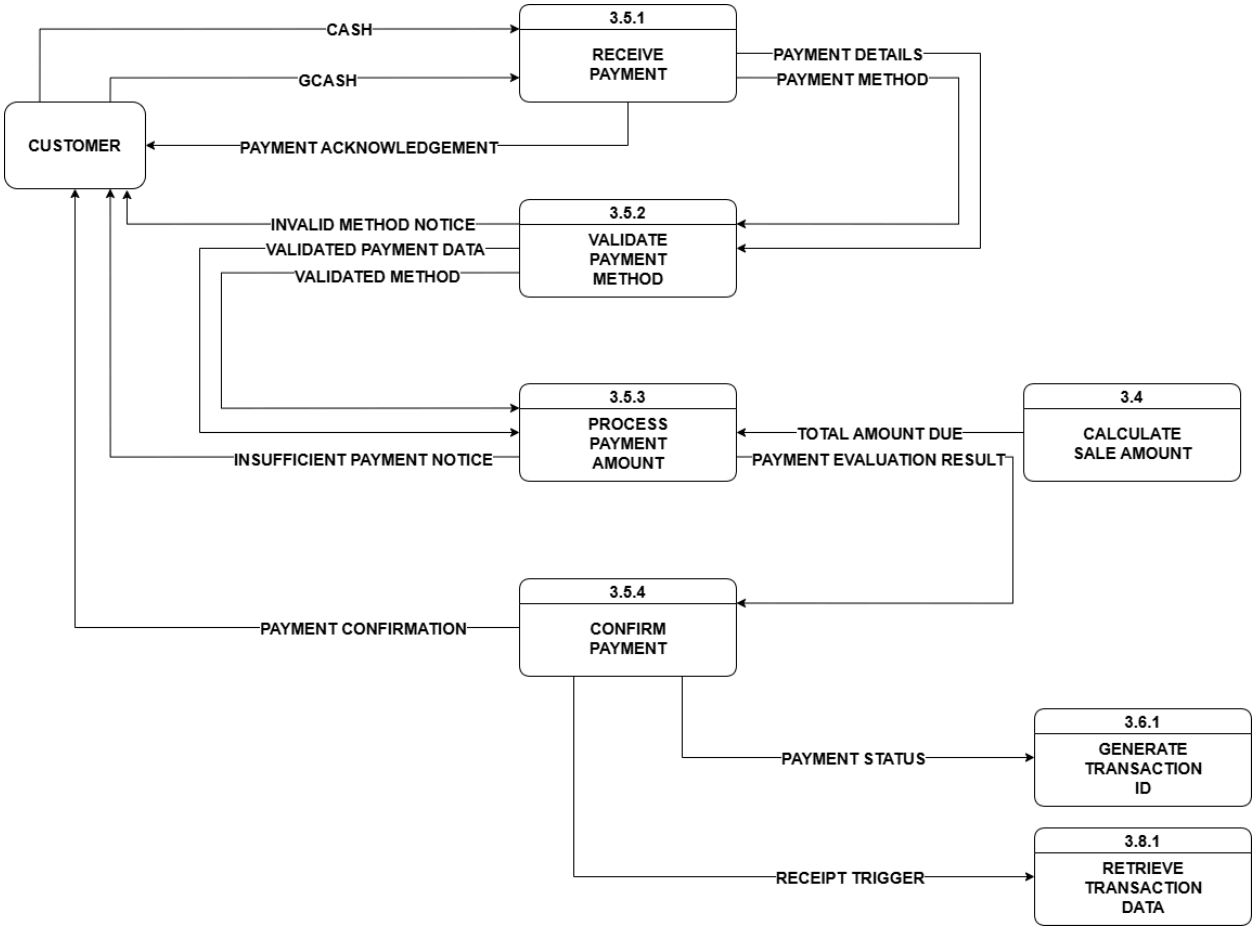
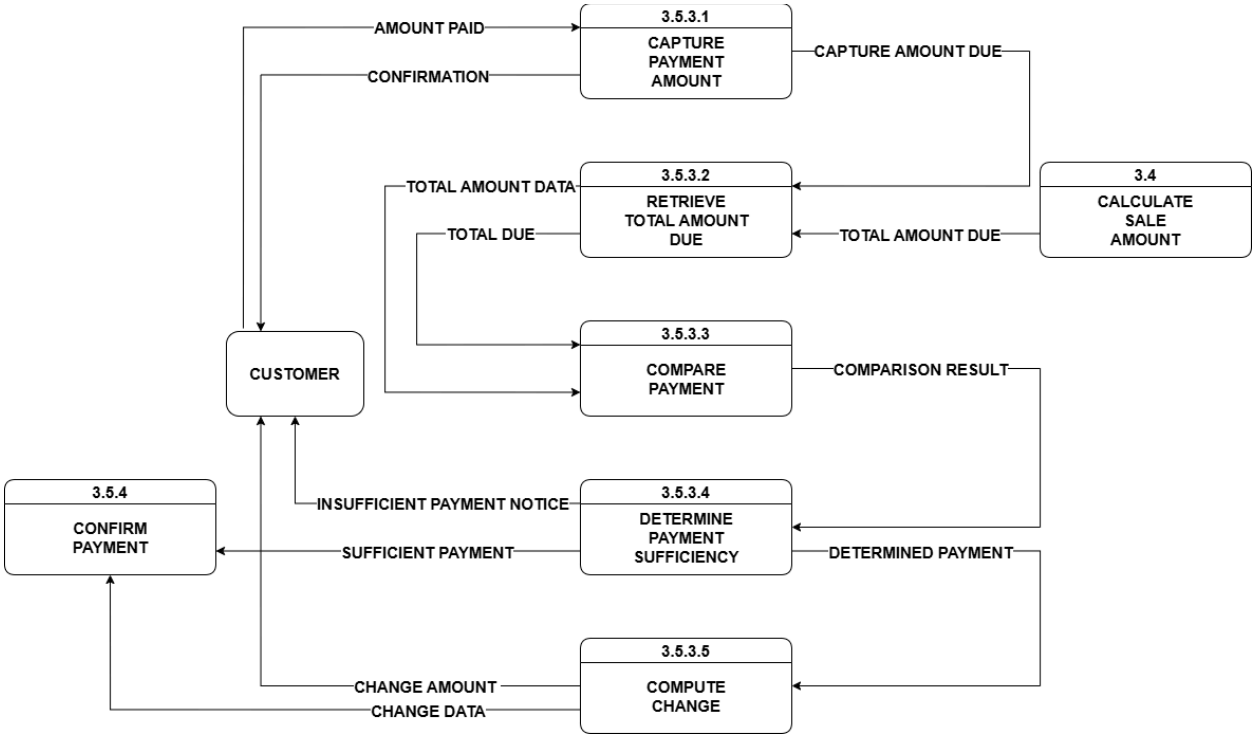


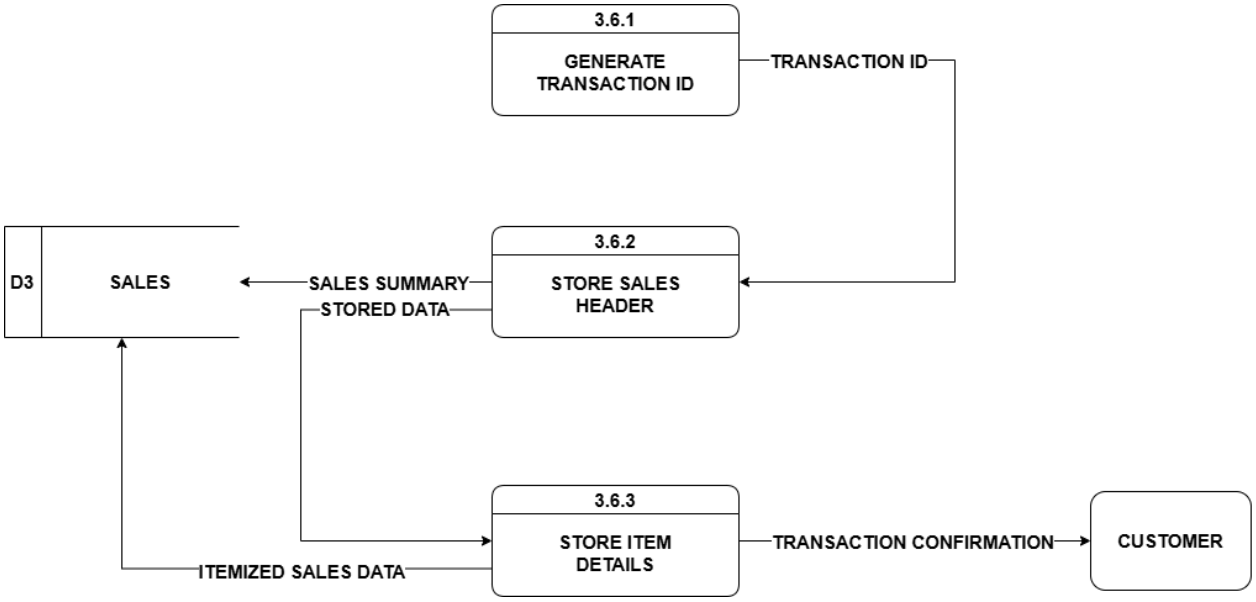
LVL 3 (3.4 COMPUTE TOTAL AMOUNT)



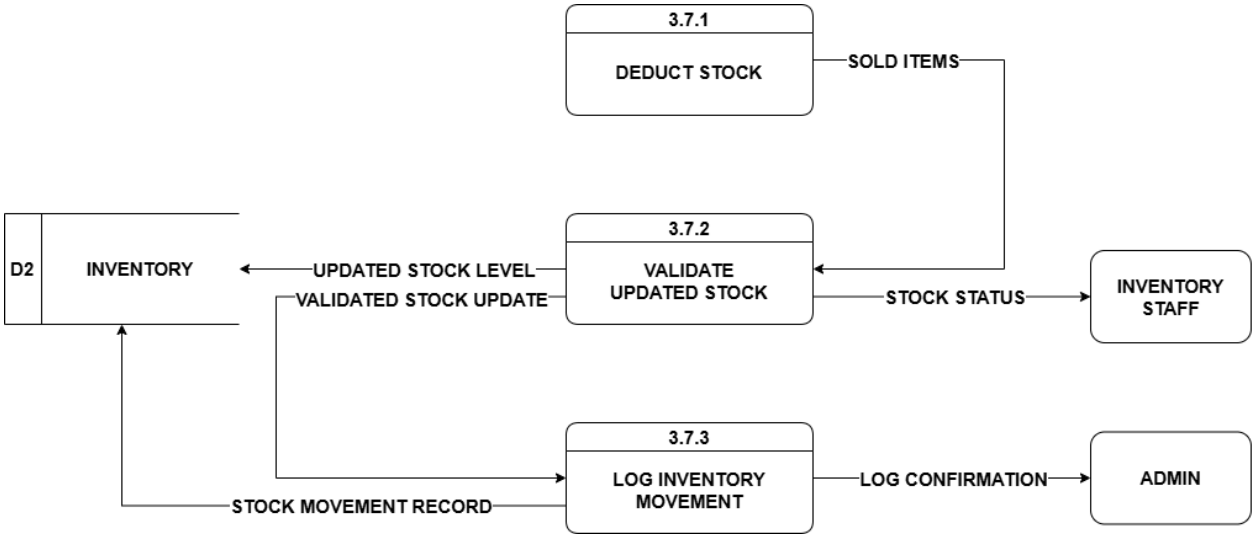
LVL 3 (3.5 PROCESS PAYMENT)



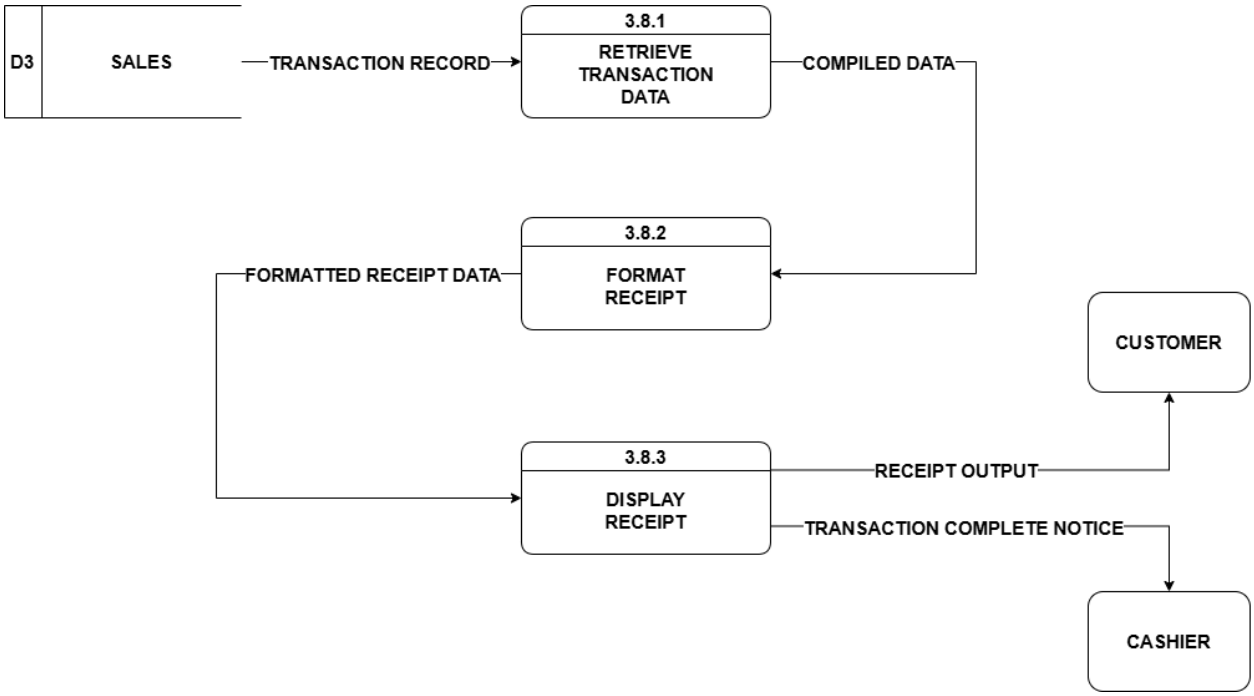
LVL 4 (3.5.3 PROCESS PAYMENT AMOUNT)



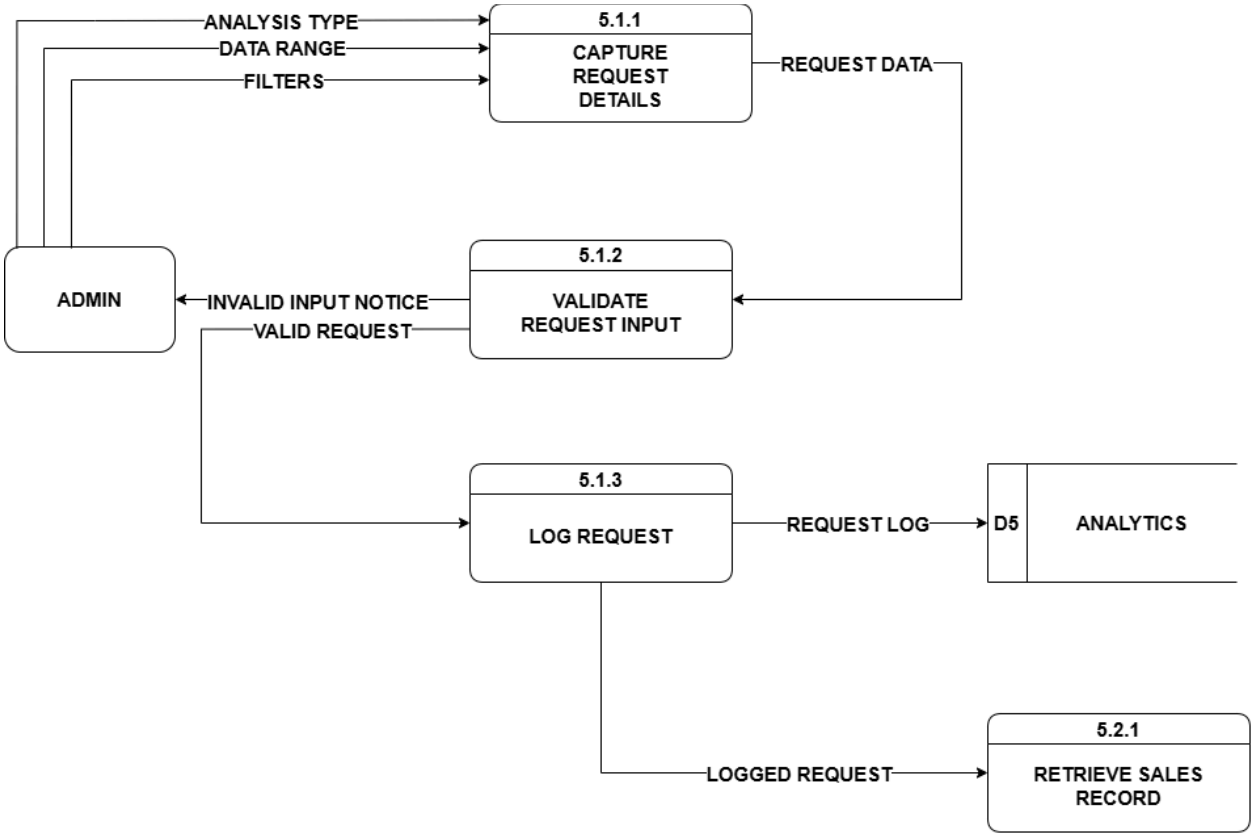
LVL 3 (3.6 RECORD TRANSACTION)



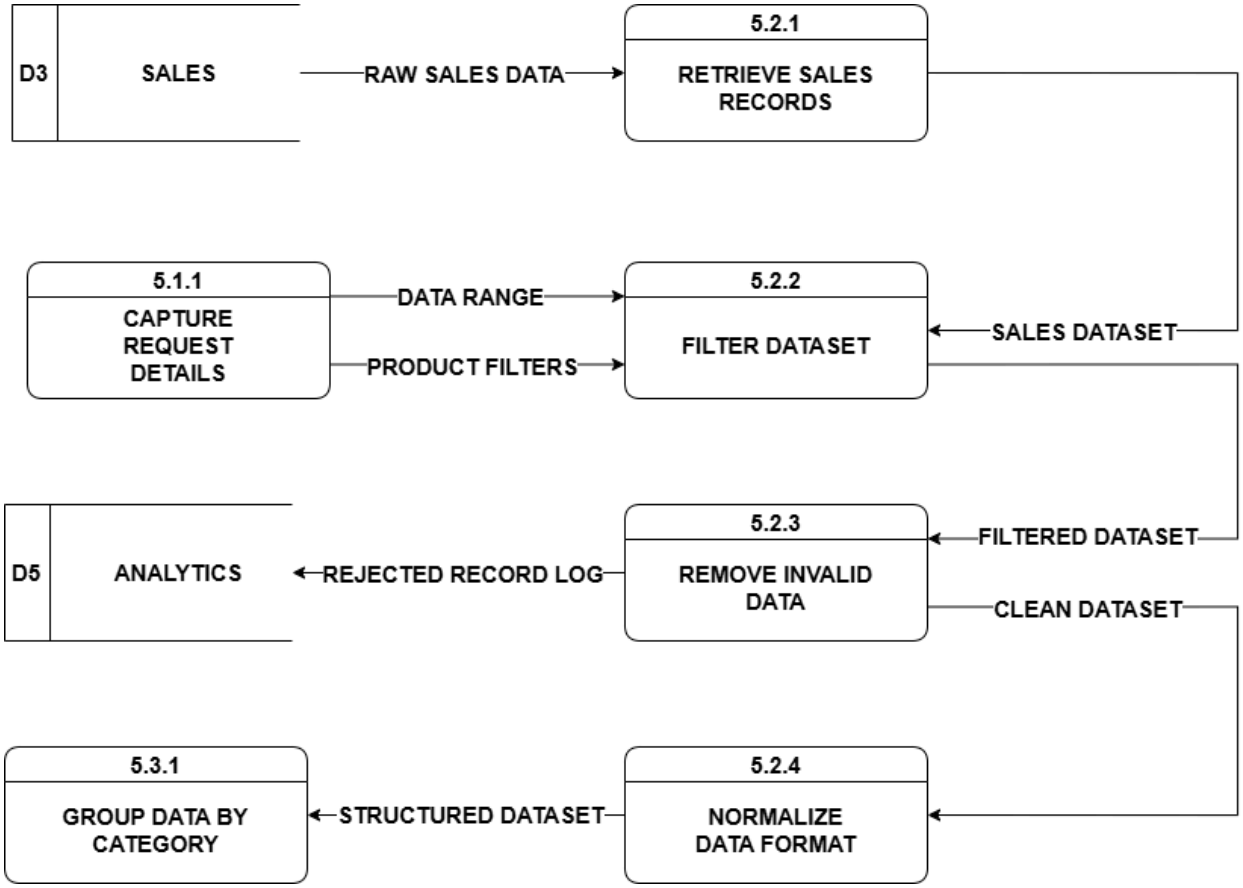
LVL 3 (UPDATE INVENTORY)



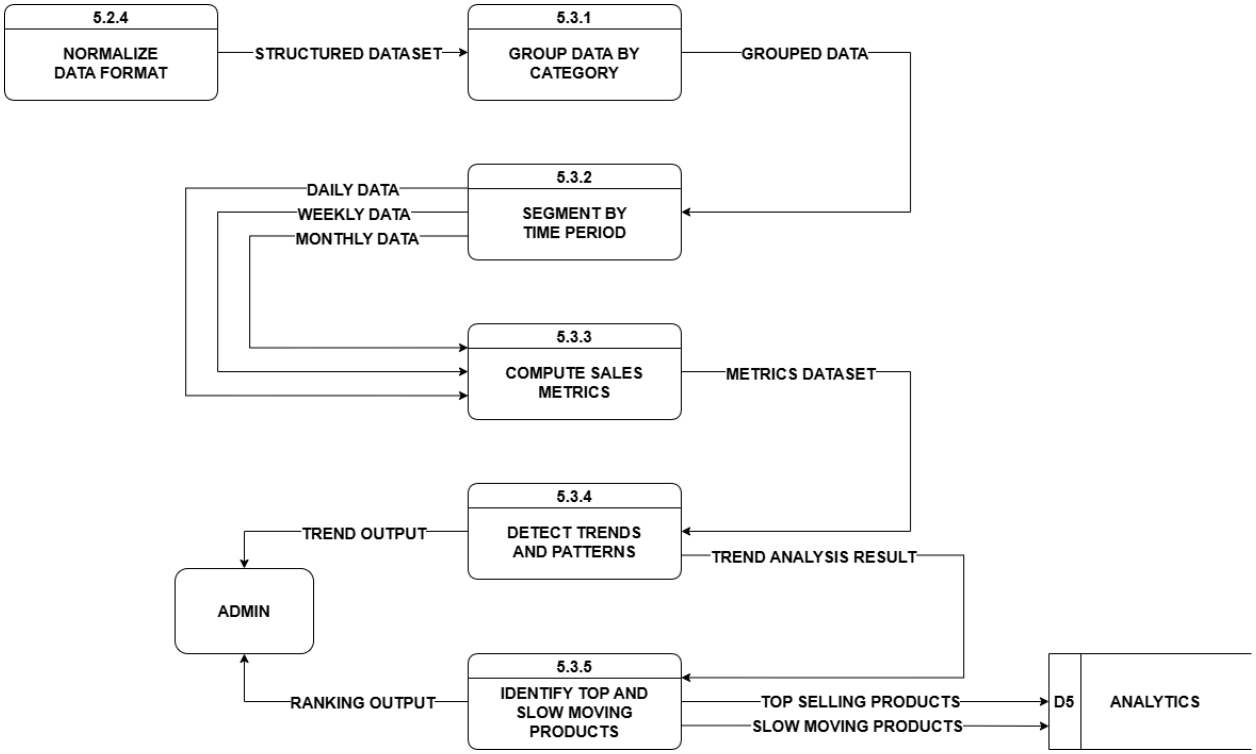
LVL 3 (3.8 GENERATE RECEIPT)



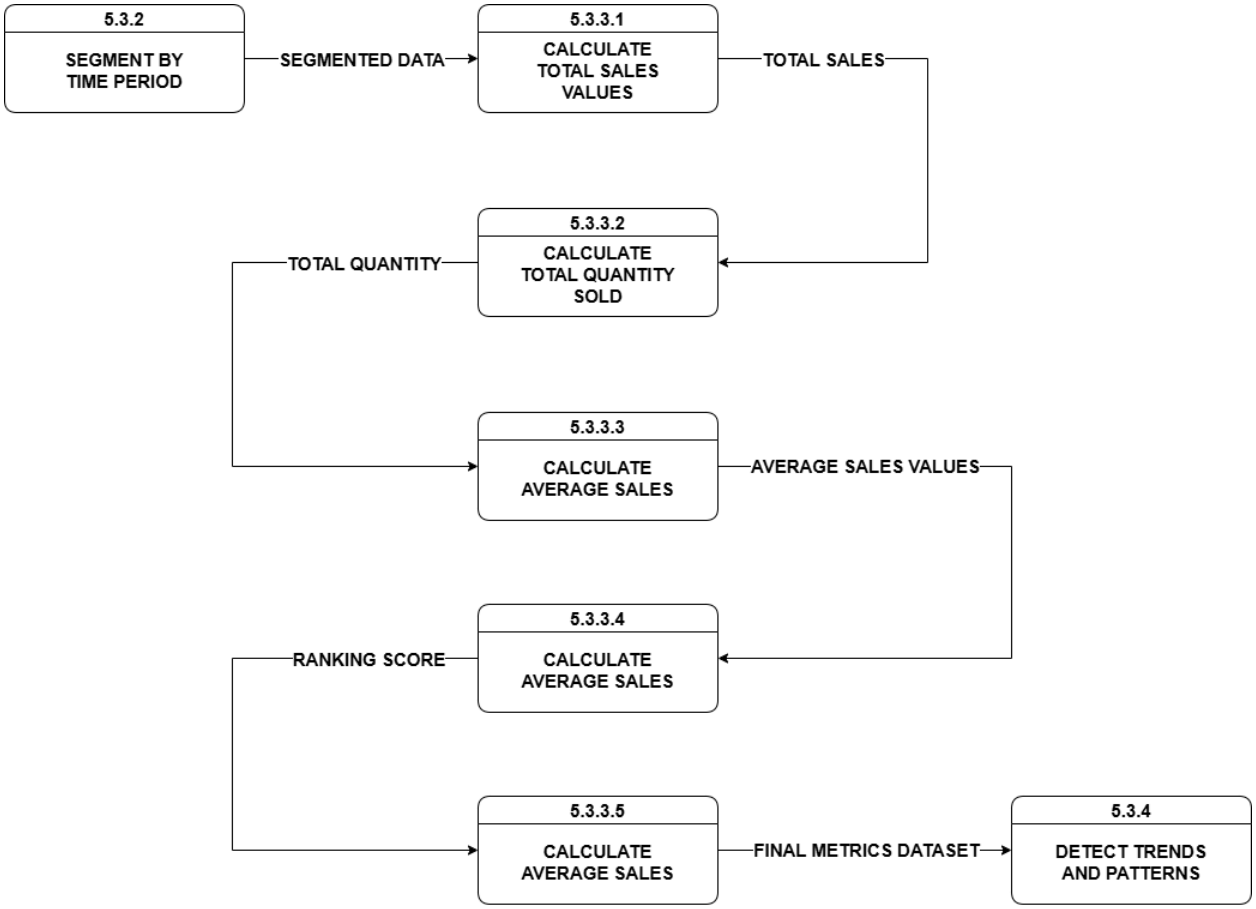
LVL 3 (5.1 RECEIVE ANALYSIS REQUEST)



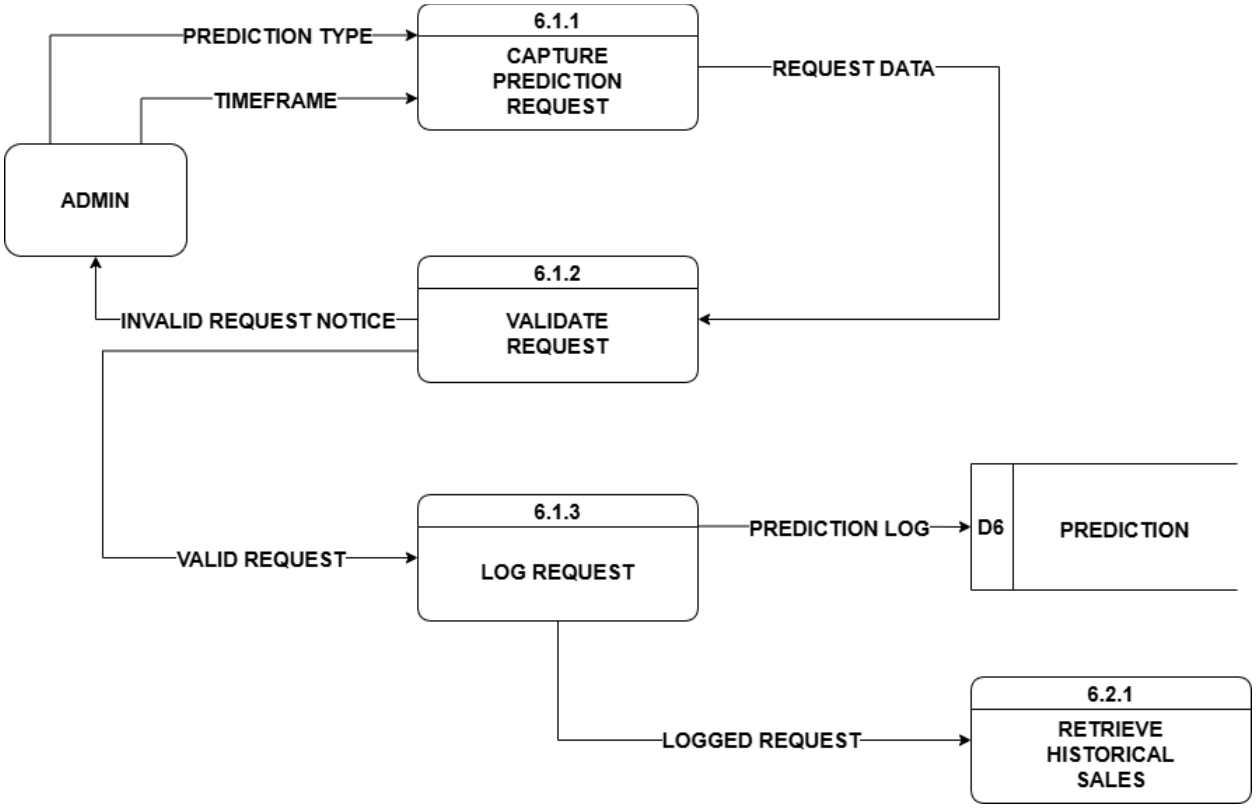
LVL 3 (5.2 PREPARE DATASET)



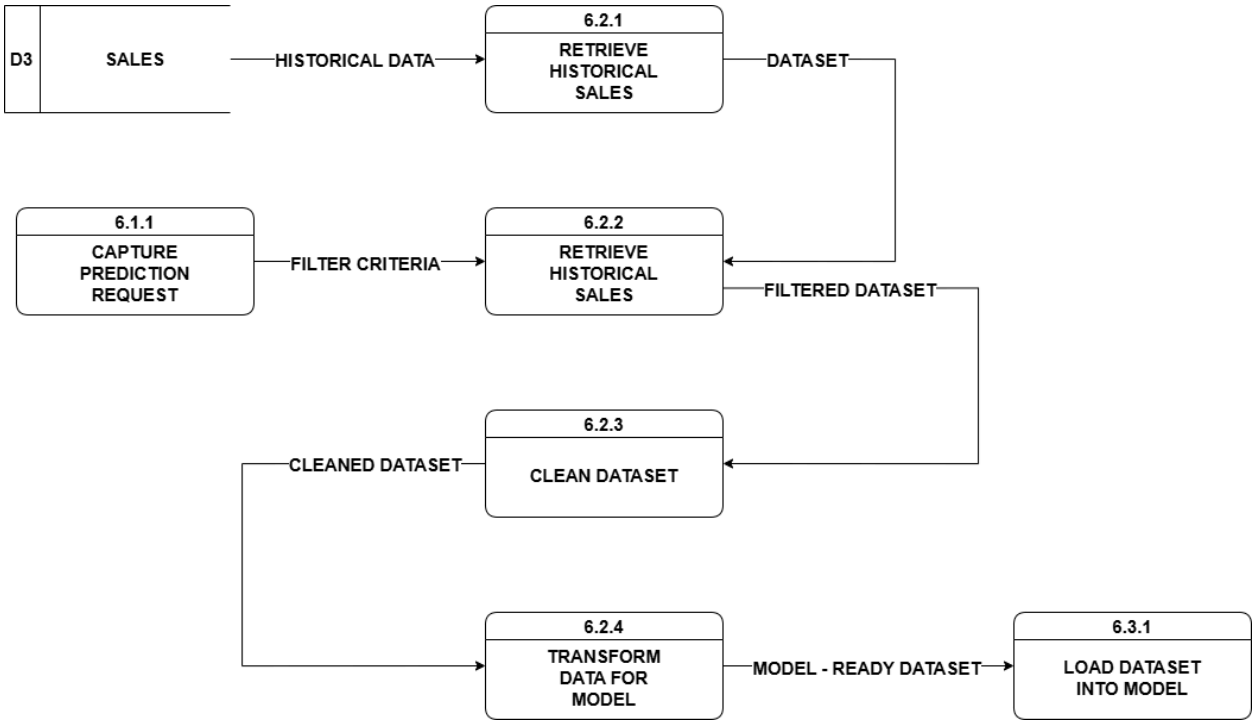
LVL 3 (5.3 ANALYZE SALES TREND)



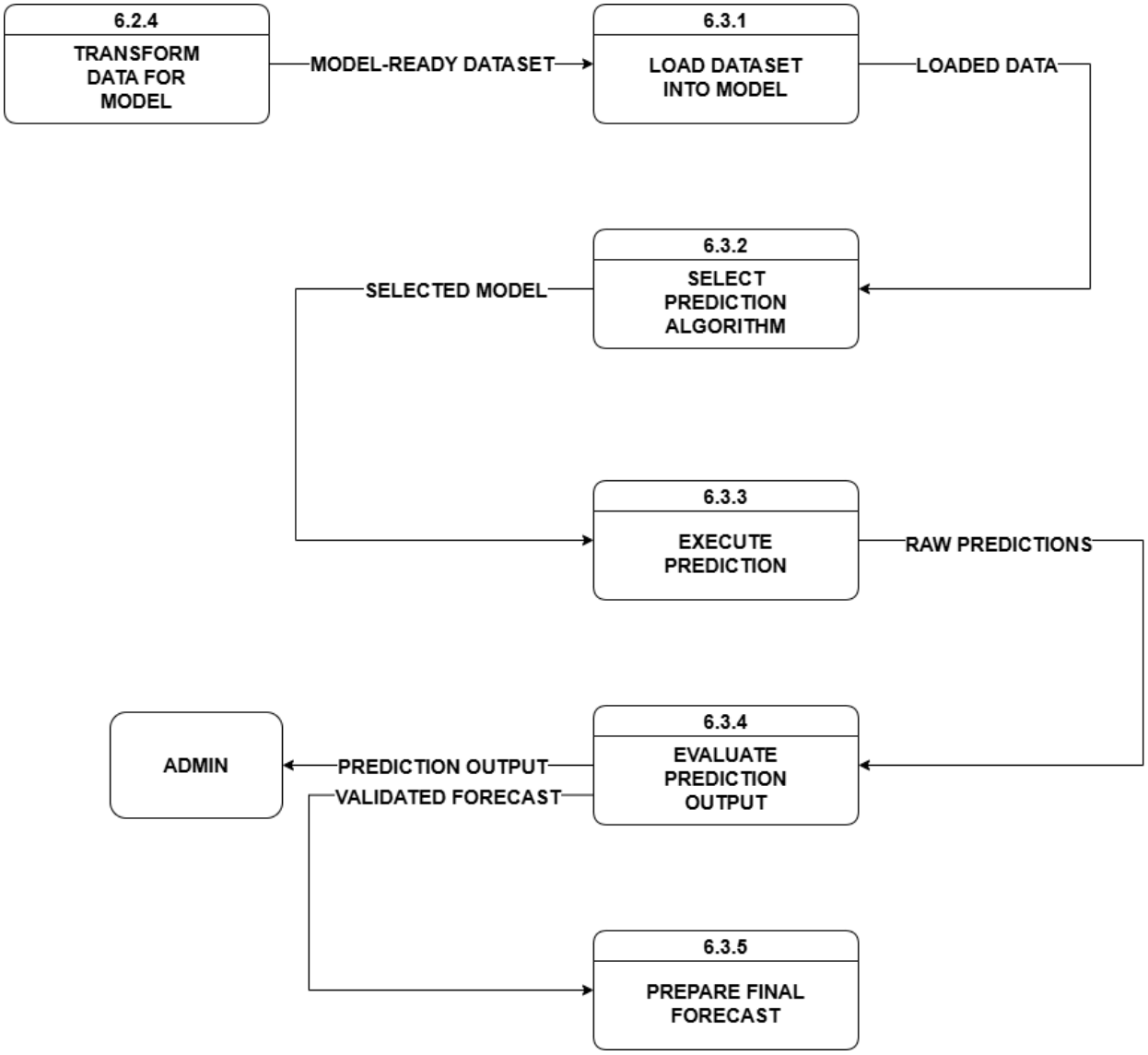
LVL 4 (5.3.3 COMPUTE METRICS)



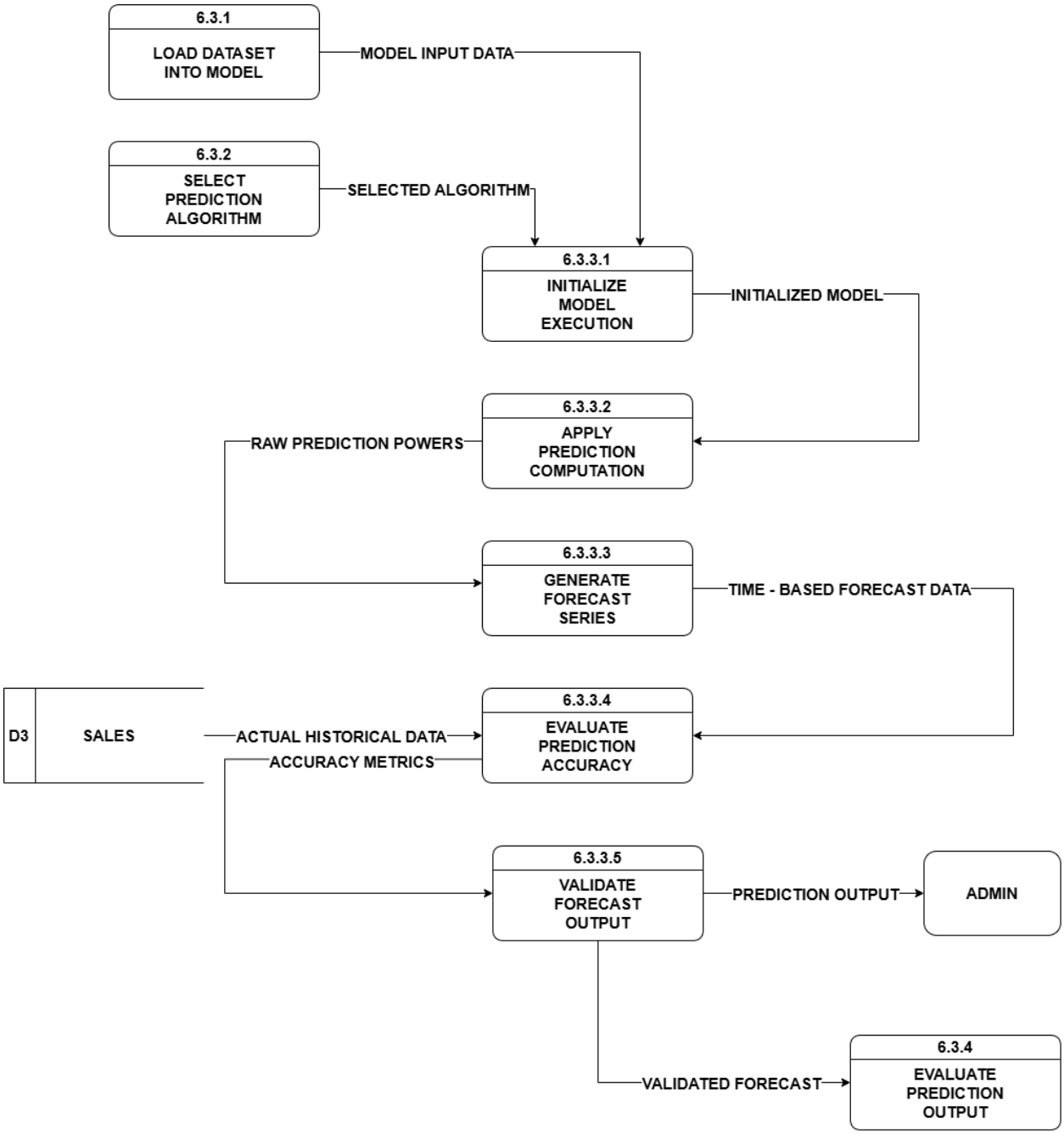
LVL 3 (6.1 RECEIVE PREDICTION REQUEST)



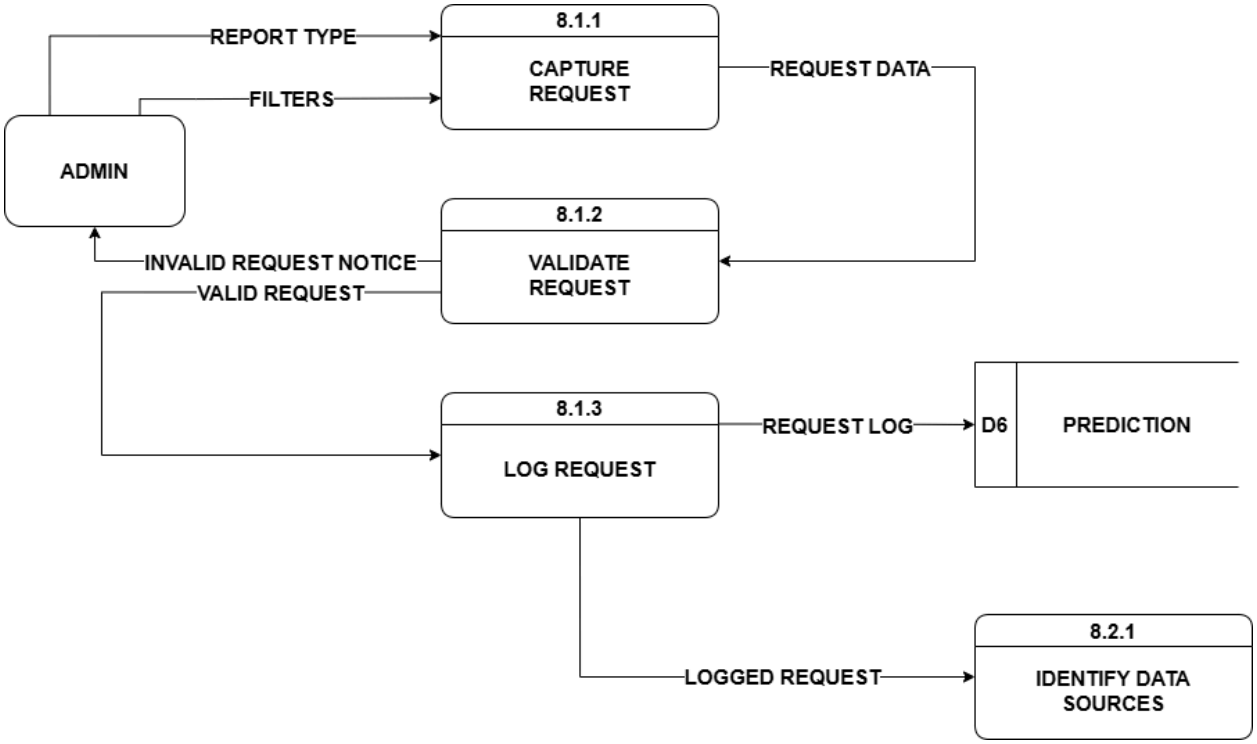
LVL 3 (6.2 PREPARE DATASET)



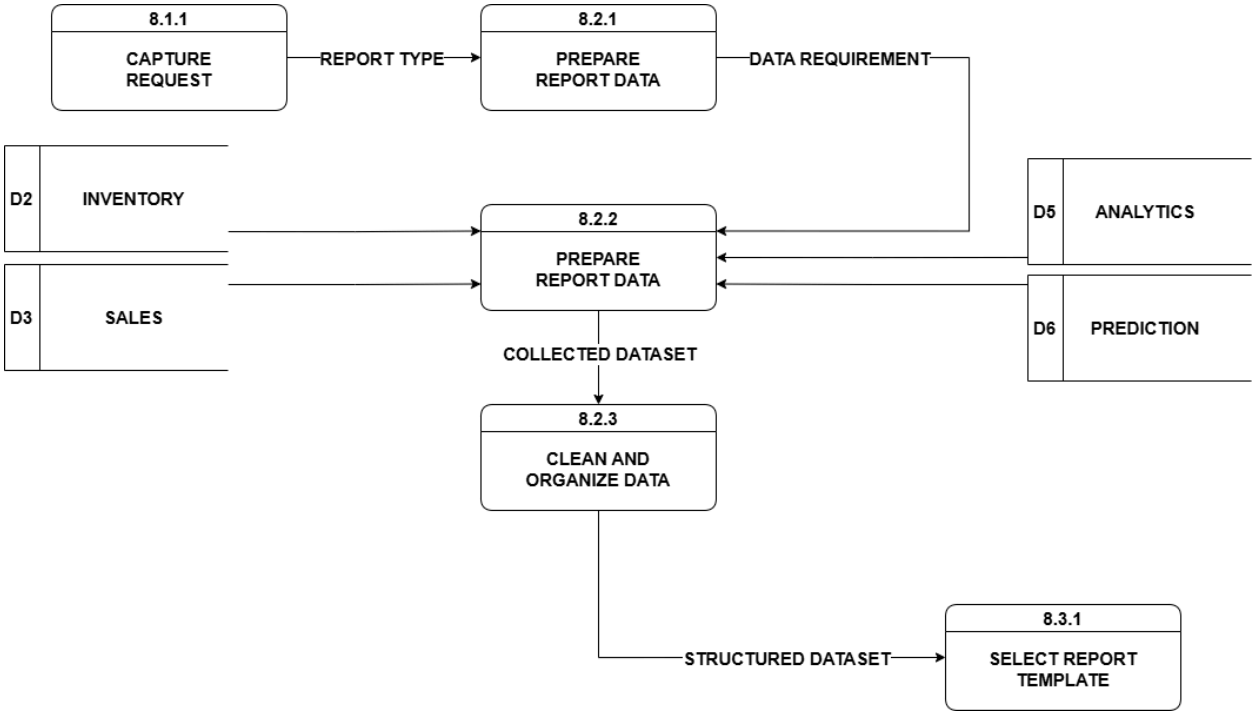
LVL 3 (6.3 RUN PREDICTION MODEL)



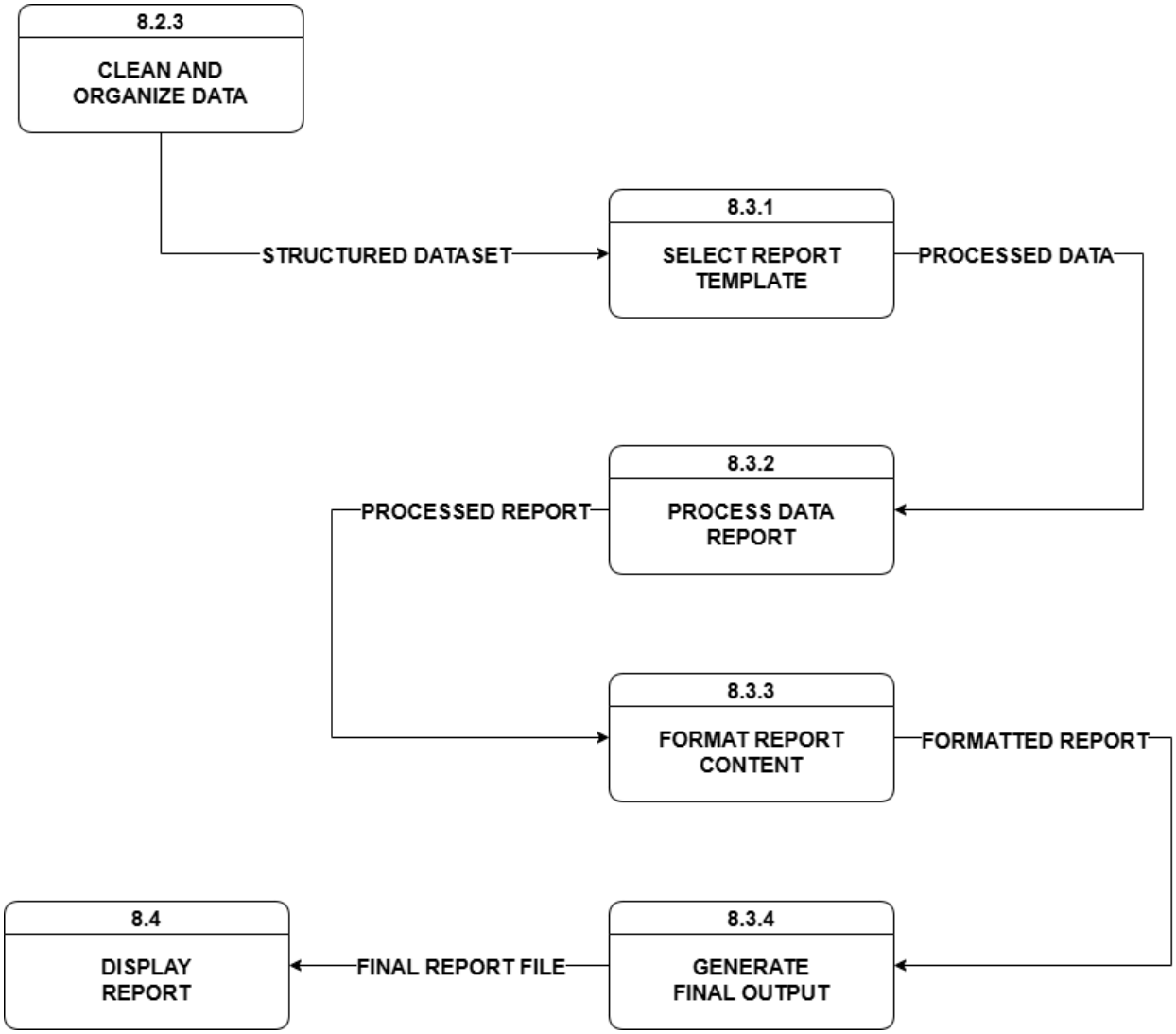
LVL 4 (6.3.3 EXECUTE PREDICTION)



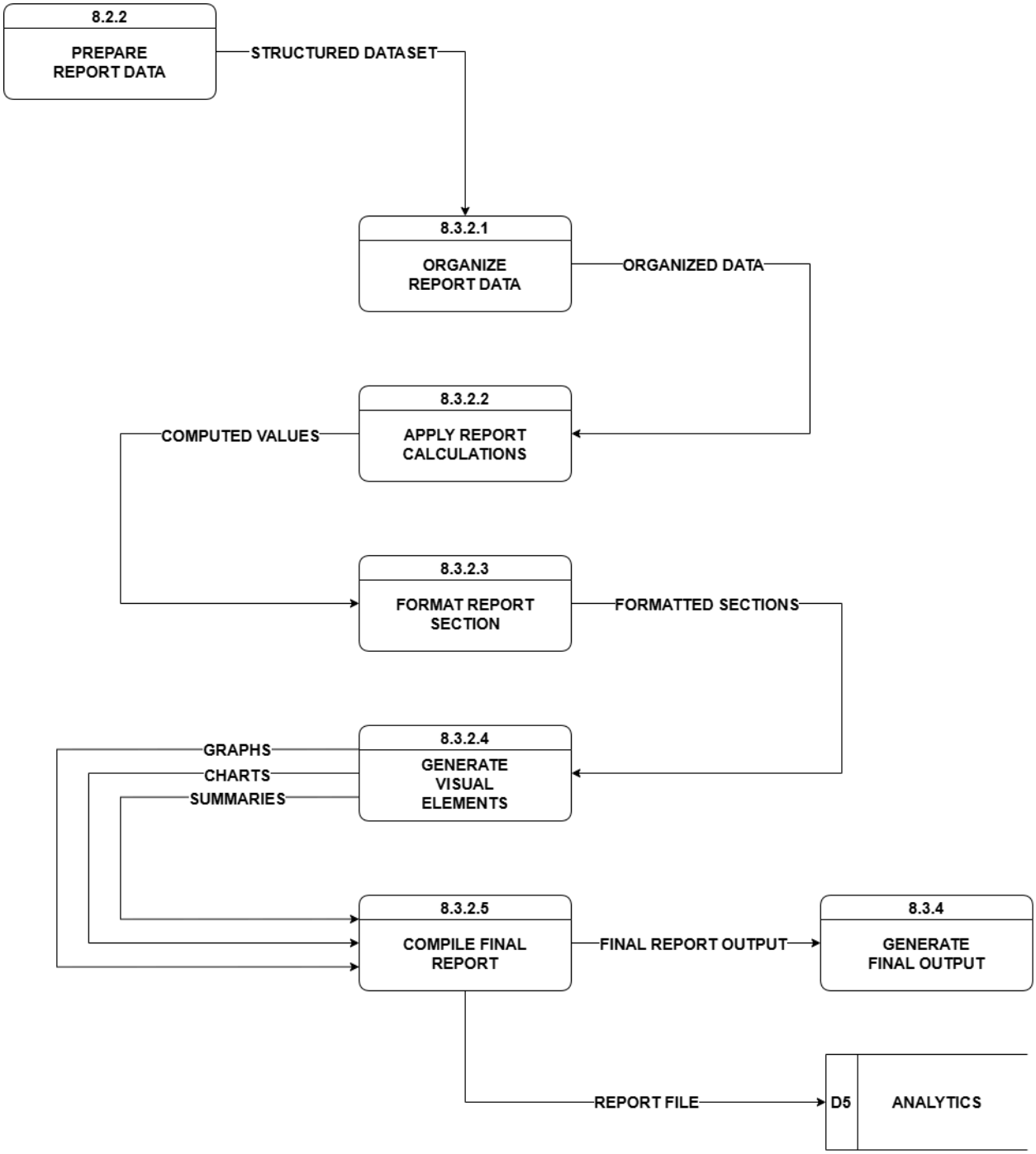
LVL 3 (8.1 RECEIVE REPORT REQUEST)



LVL 3 (8.2 PREPARE REPORT DATA LVL 3)



LVL 3 (8.3 GENERATE REPORT)



LVL 4 (8.3.2 PROCESS REPORT DATA)

3.5 Entity Relationship Diagram

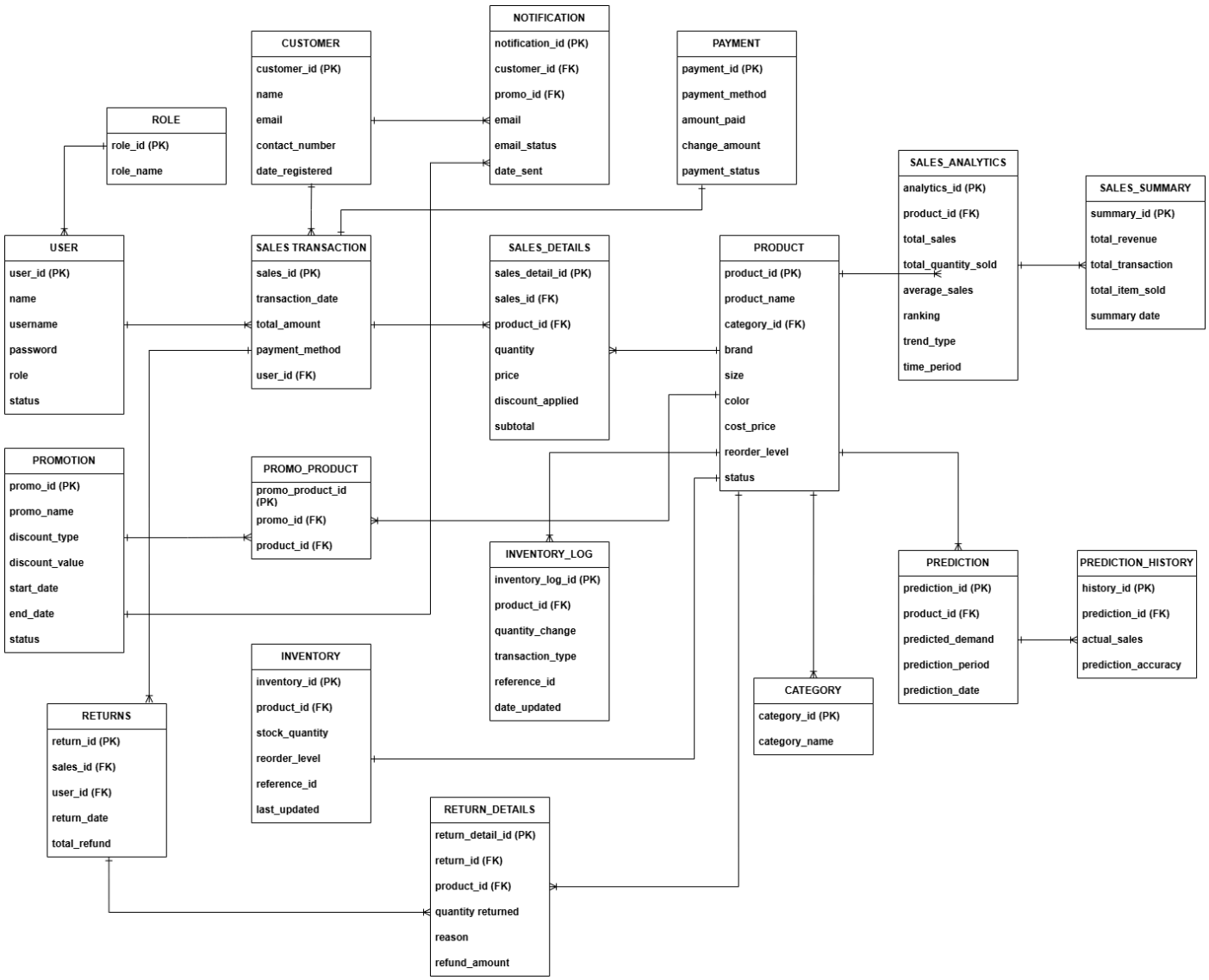


Figure 5: Entity Relationship Diagram of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing

Figure 5 presents the Entity Relationship Diagram (ERD) of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing, showing the structure and relationships among key database tables. It illustrates how

core entities such as users, customers, products, categories, sales transactions, inventory, promotions, and returns are interconnected to support daily operations. The diagram also integrates customer notifications for promotional marketing and includes analytics components such as sales analytics, summaries, and predictions to enable data-driven insights and forecasting. Overall, the ERD reflects a well-structured database that supports transaction processing, inventory management, customer engagement, and decision-making.

3.6 Use Case Diagram

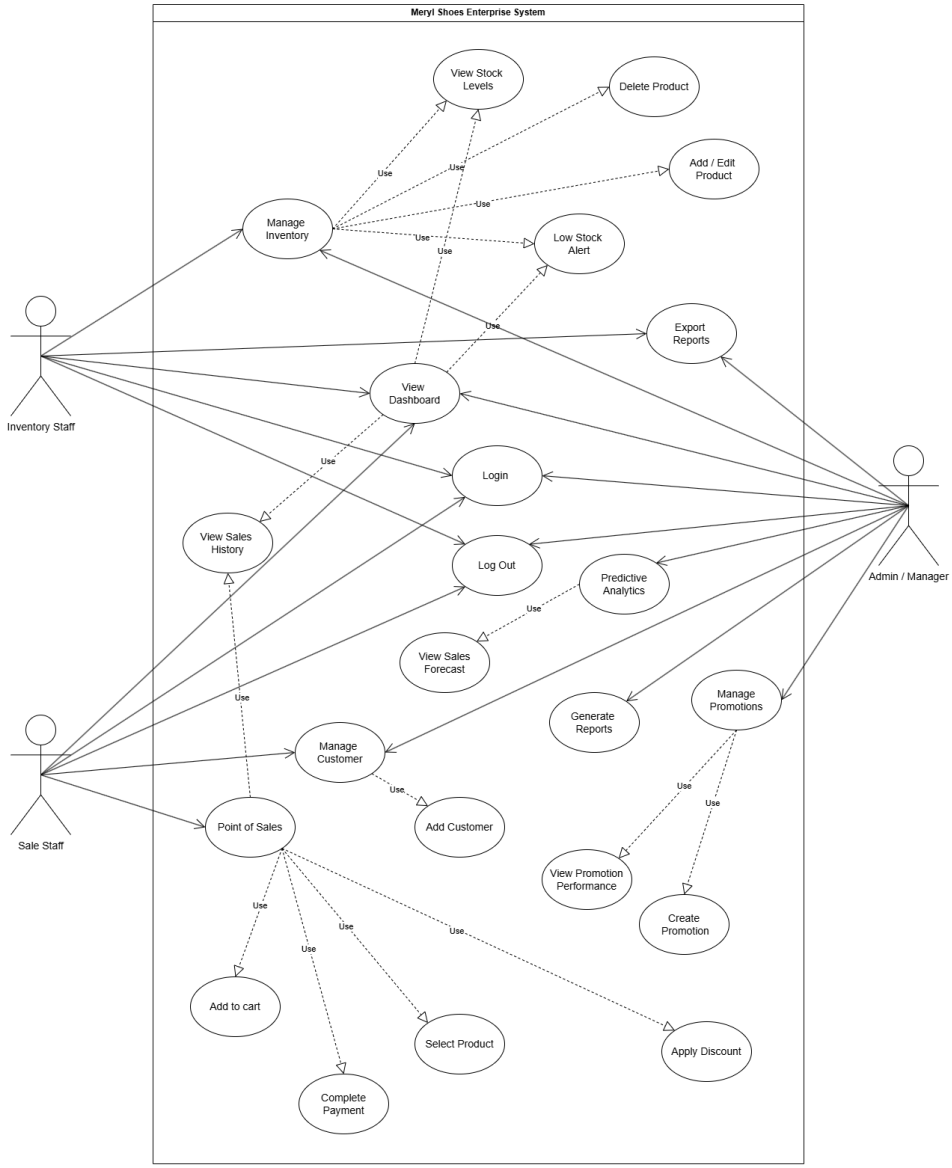


Figure 6: Use Case Diagram of the Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing

3.7 Data Dictionary

The data dictionary furnishes detailed information about the business data, like the standard deviation of data elements, their meaning, and allowable values.

Table 1

User

Field Name	Data Type	Data Format	Field Size	Description	Example
user_id	INT	Numeric	11	Unique identifier of the user	101
name	VARCHAR	Text	100	Full name of user	Juan Dela Cruz
username	VARCHAR	Text	50	Login username	admin01
password	VARCHAR	Text (Encrypted)	255	User password	$2y$10$abc123
role	VARCHAR	Text	50	User role (Admin, Cashier)	Admin
status	VARCHAR	Text	20	Account status	Active

Table 2

Role

Field Name	Data Type	Data Format	Field Size	Description	Example
role_id	INT	Numeric	11	Unique role ID	1
role_name	VARCHAR	Text	50	Role name	Admin

Table 3

Customer

Field Name	Data Type	Data Format	Field Size	Description	Example
customer_id	INT	Numeric	11	Unique customer ID	201
name	VARCHAR	Text	100	Customer name	Maria Santos
email	VARCHAR	Email	100	Customer email	maria@gmail.com
contact_number	VARCHAR	Numeric/Text	15	Contact number	09123456789
date_registered	DATE	YYYY-MM-DD	—	Registration date	2026-04-01

Table 4

Notification

Field Name	Data Type	Data Format	Field Size	Description	Example
notification_id	INT	Numeric	11	Unique notification ID	301
customer_id	INT	Numeric	11	Linked customer	201
promo_id	INT	Numeric	11	Linked promotion	401
email	VARCHAR	Email	100	Email sent to	maria@gmail.com
email_status	VARCHAR	Text	20	Status of email	Sent
date_sent	DATETIME	YYYY-MM-DD HH:MM	—	Date sent	2026-04-10 10:30

Table 5

Sales_Transaction

Field Name	Data Type	Data Format	Field Size	Description	Example
sales_id	INT	Numeric	11	Unique transaction ID	5001
transaction_date	DATETIME	YYYY-MM-DD HH:MM	—	Date of transaction	2026-04-10 14:20
total_amount	DECIMAL	10,2	—	Total sale amount	2500.00
payment_method	VARCHAR	Text	50	Mode of payment	Cash
user_id	INT	Numeric	11	Cashier ID	101

Table 6

Sales_Details

Field Name	Data Type	Data Format	Field Size	Description	Example
sales_detail_id	INT	Numeric	11	Unique sales detail ID	6001
sales_id	INT	Numeric	11	Linked transaction	5001
product_id	INT	Numeric	11	Product sold	301
quantity	INT	Numeric	11	Quantity sold	2
price	DECIMAL	10,2	—	Product price	1200.00
discount_applied	DECIMAL	10,2	—	Discount value	100.00
subtotal	DECIMAL	10,2	—	Total per item	2300.00

Table 7

Product

Field Name	Data Type	Data Format	Field Size	Description	Example
product_id	INT	Numeric	11	Unique product ID	301
product_name	VARCHAR	Text	100	Product name	Nike Air Max
category_id	INT	Numeric	11	Product category	10
brand	VARCHAR	Text	50	Brand name	Nike
size	VARCHAR	Text	10	Product size	42
color	VARCHAR	Text	30	Product color	Black
cost_price	DECIMAL	10,2	—	Cost price	1500.00
reorder_level	INT	Numeric	11	Reorder threshold	5
status	VARCHAR	Text	20	Product status	Available

Table 8

Inventory

Field Name	Data Type	Data Format	Field Size	Description	Example
inventory_id	INT	Numeric	11	Unique inventory ID	7001
product_id	INT	Numeric	11	Linked product	301
stock_quantity	INT	Numeric	11	Current stock	50
reorder_level	INT	Numeric	11	Minimum stock level	5
reference_id	INT	Numeric	11	Related transaction	5001
last_updated	DATETIME	YYYY-MM-DD HH:MM	—	Last update	2026-04-10

Table 9

Inventory_Log

Field Name	Data Type	Data Format	Field Size	Description	Example
inventory_log_id	INT	Numeric	11	Log ID	8001
product_id	INT	Numeric	11	Product reference	301
quantity_change	INT	Numeric	11	Stock change	-2
transaction_type	VARCHAR	Text	50	Type (Sale/Return)	Sale
reference_id	INT	Numeric	11	Related transaction	5001
date_updated	DATETIME	YYYY-MM-DD HH:MM	—	Update time	2026-04-10

Table 10

Promotion

Field Name	Data Type	Data Format	Field Size	Description	Example
promo_id	INT	Numeric	11	Promotion ID	401
promo_name	VARCHAR	Text	100	Promo name	Summer Sale
discount_type	VARCHAR	Text	20	Type (Percent/Fixed)	Percent
discount_value	DECIMAL	5,2	—	Discount value	10.00
start_date	DATE	YYYY-MM-DD	—	Start date	2026-04-01
end_date	DATE	YYYY-MM-DD	—	End date	2026-04-30
status	VARCHAR	Text	20	Promo status	Active

Table 11

Promo_Product

Field Name	Data Type	Data Format	Field Size	Description	Example
promo_product_id	INT	Numeric	11	Unique ID	9001
promo_id	INT	Numeric	11	Promotion ID	401
product_id	INT	Numeric	11	Product ID	301

Table 12

Sales_Analytics

Field Name	Data Type	Data Format	Field Size	Description	Example
analytics_id	INT	Numeric	11	Analytics ID	10001
product_id	INT	Numeric	11	Product reference	301
total_sales	DECIMAL	10,2	—	Total revenue	50000.00
total_quantity_sold	INT	Numeric	11	Total items sold	200
average_sales	DECIMAL	10,2	—	Average sales	250.00
ranking	INT	Numeric	11	Product rank	1
trend_type	VARCHAR	Text	20	Trend classification	Top
time_period	VARCHAR	Text	20	Period (Monthly)	Monthly

Table 13

Sales_Summary

Field Name	Data Type	Data Format	Field Size	Description	Example
summary_id	INT	Numeric	11	Summary ID	11001
total_revenue	DECIMAL	10,2	—	Total revenue	100000.00
total_transaction	INT	Numeric	11	Total transactions	300
total_item_sold	INT	Numeric	11	Total items sold	500
summary_date	DATE	YYYY-MM-DD	—	Date	2026-04-10

Table 14

Prediction

Field Name	Data Type	Data Format	Field Size	Description	Example
prediction_id	INT	Numeric	11	Prediction ID	12001
product_id	INT	Numeric	11	Product reference	301
predicted_demand	INT	Numeric	11	Forecasted demand	150
prediction_period	VARCHAR	Text	20	Period	Weekly
prediction_date	DATE	YYYY-MM-DD	—	Date	2026-04-15

Table 15

Prediction_History

Field Name	Data Type	Data Format	Field Size	Description	Example
prediction_id	INT	Numeric	11	Prediction ID	12001
product_id	INT	Numeric	11	Product reference	301
predicted_demand	INT	Numeric	11	Forecasted demand	150
prediction_period	VARCHAR	Text	20	Period	Weekly
prediction_date	DATE	YYYY-MM-DD	—	Date	2026-04-15

Table 16

Returns

Field Name	Data Type	Data Format	Field Size	Description	Example
return_id	INT	Numeric	11	Unique identifier for return transaction	14001
sales_id	INT	Numeric	11	Reference to original sales transaction	5001
user_id	INT	Numeric	11	Staff who processed the return	101
return_date	DATETIME	YYYY-MM-DD HH:MM	—	Date and time of return	2026-04-12 15:30
total_refund	DECIMAL	10,2	—	Total refunded amount	1500.00

Table 17

Return_Details

Field Name	Data Type	Data Format	Field Size	Description	Example
payment_id	INT	Numeric	11	Linked return	14001
payment_method	VARCHAR	Text	50	Method	Cash
amount_paid	DECIMAL	10,2	—	Refund	500.00
change_amount	DECIMAL	10,2	—	Change	0.00
payment_status	VARCHAR	Text	20	Status	Refunded

Table 18

Category

3.8 Hardware and Software Requirements

Hardware and Software Requirements provide the detailed minimum requirements needed to support the functions and execution of the developed Integrated POS-Driven Inventory System with Data Analytics for Targeting Sales Marketing.

Software Requirements

- Operating System: Windows 10, macOS, or Linux.
- Web Server: Apache or Nginx, with hosting services such as Hostinger
- Database: Supabase for storing sales, inventory, user, and analytics data.
- Programming Languages:
  - Frontend: HTML, CSS, JavaScript (with Bootstrap 5).
  - Backend: Python
- Frameworks/Libraries: Laravel, Express.js, or other applicable frameworks for server-side development.
- Analytics Tools: Chart.js or similar libraries for data visualization and analytics reporting.
- Version Control: Git for source code management.
- API Integration: RESTful APIs for data exchange and external services.

Recommended Minimum Hardware Requirements

The following is the list of hardware devices needed to support the functions and execution of the proposed system:

- Computer: A desktop or laptop computer with internet access to host and access the system.
- Processor: A multi-core processor with a speed of at least 2.0 GHz.
- RAM: At least 8 GB of RAM for smooth system operation and multitasking.
- Storage: Minimum of 100 GB of free storage space to accommodate system files, database, reports, and user data.

- Printer: Receipt printer or standard printer for printing transaction receipts and reports.
- Network: Stable internet connection with at least 5 Mbps download speed for optimal performance and data access.

3.9 Use Case Scenario

Table 1
Use Case 1: Login
Use Case Goal: To allow authorized users to access the Meryl Shoes Enterprise System.

Scenario 1: Administrator logs into the system.
Actor: Administrator
Pre-condition: The Administrator must have a valid account registered in the system.
Main Flow:
Step 1: The Administrator opens the login page.
Step 2: The Administrator enters the username and password.
Step 3: The Administrator clicks the Login button.
Step 4: The system validates the entered credentials.
User Action: Administrator enters valid login credentials and clicks the Login button.
System Response: The system grants access and redirects the Administrator to the dashboard.

Scenario 2: Inventory Staff logs into the system.
Actor: Inventory Staff
Pre-condition: The Inventory Staff must have a valid account registered in the system.
Main Flow:
Step 1: The Inventory Staff opens the login page.
Step 2: The Inventory Staff enters the username and password.

Step 3: The Inventory Staff clicks the Login button.
Step 4: The system validates the entered credentials.
User Action: Inventory Staff enters valid login credentials and clicks the Login button.
System Response: The system grants access and redirects the Inventory Staff to the dashboard.

Scenario 3: Sales Staff logs into the system.
Actor: Sales Staff
Pre-condition: The Sales Staff must have a valid account registered in the system.
Main Flow:
Step 1: The Sales Staff opens the login page.
Step 2: The Sales Staff enters the username and password.
Step 3: The Sales Staff clicks the Login button.
Step 4: The system validates the entered credentials.
User Action: Sales Staff enters valid login credentials and clicks the Login button.
System Response: The system grants access and redirects the Sales Staff to the dashboard.

Table 2
Use Case 2: View Dashboard
Use Case Goal: To allow users to view the summary of business activities.

Scenario 1: Administrator views the dashboard.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system.
Main Flow:
Step 1: The Administrator logs into the system.
Step 2: The Administrator opens the dashboard page.
Step 3: The system retrieves the latest business data.
Step 4: The Administrator reviews the displayed information.
User Action: Administrator clicks the Dashboard menu.

System Response: The system displays revenue, stock status, customer count, low stock items, and top-selling products.

Scenario 2: Inventory Staff views the dashboard.
Actor: Inventory Staff
Pre-condition: The Inventory Staff must be logged into the system.
Main Flow:
Step 1: The Inventory Staff logs into the system.
Step 2: The Inventory Staff opens the dashboard page.
Step 3: The system retrieves the latest inventory-related data.
Step 4: The Inventory Staff reviews the displayed information.
User Action: Inventory Staff clicks the Dashboard menu.
System Response: The system displays stock summary, low stock items, recent sales, and top-selling products.

Scenario 3: Sales Staff views the dashboard.
Actor: Sales Staff
Pre-condition: The Sales Staff must be logged into the system.
Main Flow:
Step 1: The Sales Staff logs into the system.
Step 2: The Sales Staff opens the dashboard page.
Step 3: The system retrieves the latest sales-related data.
Step 4: The Sales Staff reviews the displayed information.
User Action: Sales Staff clicks the Dashboard menu.
System Response: The system displays total revenue, products in stock, total customers, low stock items, recent sales, and top-selling products.

Table 3
Use Case 3: Manage Inventory
Use Case Goal: To allow authorized users to manage and monitor inventory records.

Scenario 1: Inventory Staff accesses inventory management.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and authorized to access inventory.

Main Flow:

Step 1: The Inventory Staff logs into the system.

Step 2: The Inventory Staff opens the Inventory Management page.

Step 3: The system displays the list of inventory items.

Step 4: The Inventory Staff reviews the available inventory records.

User Action: Inventory Staff clicks the Inventory Management menu.

System Response: The system displays the inventory list with product details.

Scenario 2: Administrator accesses inventory management.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and authorized to access inventory.

Main Flow:

Step 1: The Administrator logs into the system.

Step 2: The Administrator opens the Inventory Management page.

Step 3: The system displays the list of inventory items.

Step 4: The Administrator reviews the available inventory records.

User Action: Administrator clicks the Inventory Management menu.

System Response: The system displays the inventory list with product details.

Scenario 3: Inventory Staff searches for an inventory item.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and currently viewing the inventory page.

Main Flow:

Step 1: The Inventory Staff opens the Inventory Management page.

Step 2: The Inventory Staff enters a keyword in the search field.

Step 3: The system filters the inventory records.

Step 4: The Inventory Staff reviews the filtered inventory results.

User Action: Inventory Staff enters a keyword in the Search field.

System Response: The system updates the inventory list based on the search term.

Scenario 4: Administrator searches for an inventory item.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and currently viewing the inventory page.

Main Flow:

Step 1: The Administrator opens the Inventory Management page.

Step 2: The Administrator enters a keyword in the search field.

Step 3: The system filters the inventory records.

Step 4: The Administrator reviews the filtered inventory results.

User Action: Administrator enters a keyword in the Search field.

System Response: The system updates the inventory list based on the search term.

Scenario 5: Inventory Staff views stock levels.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and inventory records must exist.

Main Flow:

Step 1: The Inventory Staff opens the Inventory Management page.

Step 2: The Inventory Staff reviews the stock column of the listed products.

Step 3: The system retrieves the current quantity of each product.

User Action: Inventory Staff selects the inventory list or stock view.

System Response: The system displays the stock levels of all available items.

Scenario 6: Administrator views stock levels.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and inventory records must exist.

Main Flow:

Step 1: The Administrator opens the Inventory Management page.

Step 2: The Administrator reviews the stock column of the listed products.

Step 3: The system retrieves the current quantity of each product.

User Action: Administrator selects the inventory list or stock view.

System Response: The system displays the stock levels of all available items.

Scenario 7: Inventory Staff adds a product.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and authorized to add products.

Main Flow:

Step 1: The Inventory Staff opens the Inventory Management page.

Step 2: The Inventory Staff clicks Add Product.

Step 3: The Inventory Staff enters the product details.

Step 4: The Inventory Staff saves the new record.

User Action: Inventory Staff fills out the product form and clicks Save.

System Response: The system stores the new product and updates the inventory list.

Scenario 8: Administrator adds a product.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and authorized to add products.

Main Flow:

Step 1: The Administrator opens the Inventory Management page.

Step 2: The Administrator clicks Add Product.

Step 3: The Administrator enters the product details.

Step 4: The Administrator saves the new record.

User Action: Administrator fills out the product form and clicks Save.

System Response: The system stores the new product and updates the inventory list.

Scenario 9: Inventory Staff edits a product.
Actor: Inventory Staff
Pre-condition: The Inventory Staff must be logged into the system, authorized to edit products, and the product record must exist.
Main Flow:
Step 1: The Inventory Staff opens the Inventory Management page.
Step 2: The Inventory Staff selects a product from the list.
Step 3: The Inventory Staff updates the product details.
Step 4: The Inventory Staff saves the changes.
User Action: Inventory Staff clicks the Edit button of a selected product and updates the record.
System Response: The system saves the changes and refreshes the inventory list.

Scenario 10: Administrator edits a product.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system, authorized to edit products, and the product record must exist.
Main Flow:
Step 1: The Administrator opens the Inventory Management page.
Step 2: The Administrator selects a product from the list.
Step 3: The Administrator updates the product details.
Step 4: The Administrator saves the changes.
User Action: Administrator clicks the Edit button of a selected product and updates the record.
System Response: The system saves the changes and refreshes the inventory list.

Scenario 11: Inventory Staff deletes a product.
Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system, authorized to delete products, and the product record must exist.

Main Flow:

Step 1: The Inventory Staff opens the Inventory Management page.

Step 2: The Inventory Staff selects a product from the list.

Step 3: The Inventory Staff clicks Delete.

Step 4: The system asks for confirmation.

Step 5: The Inventory Staff confirms the deletion.

User Action: Inventory Staff selects a product and confirms the delete action.

System Response: The system removes the selected product from the inventory list.

Scenario 12: Administrator deletes a product.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system, authorized to delete products, and the product record must exist.

Main Flow:

Step 1: The Administrator opens the Inventory Management page.

Step 2: The Administrator selects a product from the list.

Step 3: The Administrator clicks Delete.

Step 4: The system asks for confirmation.

Step 5: The Administrator confirms the deletion.

User Action: Administrator selects a product and confirms the delete action.

System Response: The system removes the selected product from the inventory list.

Scenario 13: Inventory Staff receives a low stock alert.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and at least one product has reached the minimum stock threshold.

Main Flow:

Step 1: The Inventory Staff opens the dashboard or inventory page.

Step 2: The system checks the stock quantity of all products.

Step 3: The system identifies products with low stock.
Step 4: The Inventory Staff reviews the low stock notification.
User Action: Inventory Staff views the dashboard or inventory page.
System Response: The system displays low stock alerts for products that need replenishment.

Scenario 14: Administrator receives a low stock alert.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system and at least one product has reached the minimum stock threshold.
Main Flow:
Step 1: The Administrator opens the dashboard or inventory page.
Step 2: The system checks the stock quantity of all products.
Step 3: The system identifies products with low stock.
Step 4: The Administrator reviews the low stock notification.
User Action: Administrator views the dashboard or inventory page.
System Response: The system displays low stock alerts for products that need replenishment.

Table 4
Use Case 4: Point of Sale
Use Case Goal: To allow the Sales Staff to process customer purchases and complete sales transactions.

Scenario 1: Sales Staff opens the Point of Sale page.
Actor: Sales Staff
Pre-condition: The Sales Staff must be logged into the system.
Main Flow:
Step 1: The Sales Staff logs into the system.
Step 2: The Sales Staff opens the Point of Sale page.
Step 3: The system loads the product selection and transaction summary panel.

User Action: Sales Staff clicks the Point of Sale menu.
System Response: The system displays the Point of Sale interface.

Scenario 2: Sales Staff selects a product.
Actor: Sales Staff
Pre-condition: The Sales Staff must be logged into the system and products must be available.
Main Flow:
Step 1: The Sales Staff opens the Point of Sale page.
Step 2: The Sales Staff clicks the product selection field.
Step 3: The Sales Staff chooses a product from the list.
Step 4: The system loads the selected product information.
User Action: Sales Staff selects a product from the dropdown list.
System Response: The system displays the selected product in the transaction form.

Scenario 3: Sales Staff adds item to cart.
Actor: Sales Staff
Pre-condition: The Sales Staff must be logged into the system and a product must be selected.
Main Flow:
Step 1: The Sales Staff selects a product.
Step 2: The Sales Staff enters the quantity.
Step 3: The Sales Staff enters the discount if applicable.
Step 4: The Sales Staff clicks Add to Cart.
Step 5: The system calculates the subtotal.
User Action: Sales Staff enters transaction details and clicks Add to Cart.
System Response: The system adds the item to the shopping cart and updates the transaction summary.

Scenario 4: Sales Staff completes payment.
Actor: Sales Staff

Pre-condition: The Sales Staff must be logged into the system and at least one item must be in the cart.

Main Flow:

Step 1: The Sales Staff reviews the items in the cart.

Step 2: The Sales Staff enters the customer name if applicable.

Step 3: The Sales Staff selects the payment method.

Step 4: The Sales Staff clicks Complete Payment.

Step 5: The system records the transaction and updates stock quantity.

User Action: Sales Staff selects payment details and clicks Complete Payment.

System Response: The system saves the sale transaction, updates inventory, and clears the cart.

Table 5

Use Case 5: View Sales History

Use Case Goal: To allow authorized users to review recorded sales transactions.

Scenario 1: Sales Staff views sales history.

Actor: Sales Staff

Pre-condition: The Sales Staff must be logged into the system.

Main Flow:

Step 1: The Sales Staff logs into the system.

Step 2: The Sales Staff opens the Sales History page.

Step 3: The system retrieves recorded sales transactions.

Step 4: The Sales Staff reviews the sales list.

User Action: Sales Staff clicks the Sales History menu.

System Response: The system displays the list of sales records.

Scenario 2: Administrator views sales history.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system.

Main Flow:

Step 1: The Administrator logs into the system.
Step 2: The Administrator opens the Sales History page.
Step 3: The system retrieves recorded sales transactions.
Step 4: The Administrator reviews the sales list.
User Action: Administrator clicks the Sales History menu.
System Response: The system displays the list of sales records.

Scenario 3: Sales Staff searches a sales record.
Actor: Sales Staff
Pre-condition: The Sales Staff must be logged into the system and currently viewing the Sales History page.
Main Flow:
Step 1: The Sales Staff opens the Sales History page.
Step 2: The Sales Staff enters a keyword in the search field.
Step 3: The system filters the sales records.
Step 4: The Sales Staff reviews the filtered results.
User Action: Sales Staff enters an order ID, customer name, or product name in the search field.
System Response: The system displays the matching sales records.

Scenario 4: Administrator searches a sales record.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system and currently viewing the Sales History page.
Main Flow:
Step 1: The Administrator opens the Sales History page.
Step 2: The Administrator enters a keyword in the search field.
Step 3: The system filters the sales records.
Step 4: The Administrator reviews the filtered results.
User Action: Administrator enters an order ID, customer name, or product name in the search field.

System Response: The system displays the matching sales records.

Table 6

Use Case 6: Manage Customer

Use Case Goal: To allow authorized users to maintain customer records in the system.

Scenario 1: Sales Staff views customer records.

Actor: Sales Staff

Pre-condition: The Sales Staff must be logged into the system.

Main Flow:

Step 1: The Sales Staff logs into the system.

Step 2: The Sales Staff opens the Customers page.

Step 3: The system retrieves the customer records.

Step 4: The Sales Staff reviews the customer list.

User Action: Sales Staff clicks the Customers menu.

System Response: The system displays the customer directory.

Scenario 2: Administrator views customer records.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system.

Main Flow:

Step 1: The Administrator logs into the system.

Step 2: The Administrator opens the Customers page.

Step 3: The system retrieves the customer records.

Step 4: The Administrator reviews the customer list.

User Action: Administrator clicks the Customers menu.

System Response: The system displays the customer directory.

Scenario 3: Sales Staff adds a customer.

Actor: Sales Staff

Pre-condition: The Sales Staff must be logged into the system and authorized to add customers.

Main Flow:

Step 1: The Sales Staff opens the Customers page.

Step 2: The Sales Staff clicks Add Customer.

Step 3: The Sales Staff enters the customer information.

Step 4: The Sales Staff saves the record.

User Action: Sales Staff fills out the customer form and clicks Save.

System Response: The system stores the customer record and updates the customer list.

Scenario 4: Administrator adds a customer.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and authorized to add customers.

Main Flow:

Step 1: The Administrator opens the Customers page.

Step 2: The Administrator clicks Add Customer.

Step 3: The Administrator enters the customer information.

Step 4: The Administrator saves the record.

User Action: Administrator fills out the customer form and clicks Save.

System Response: The system stores the customer record and updates the customer list.

Scenario 5: Sales Staff searches a customer record.

Actor: Sales Staff

Pre-condition: The Sales Staff must be logged into the system and currently viewing the Customers page.

Main Flow:

Step 1: The Sales Staff opens the Customers page.

Step 2: The Sales Staff enters a keyword in the search field.

Step 3: The system filters the customer records.
Step 4: The Sales Staff reviews the filtered results.
User Action: Sales Staff enters a customer name, email, or phone number in the search field.
System Response: The system displays the matching customer records.

Scenario 6: Administrator searches a customer record.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system and currently viewing the Customers page.
Main Flow:
Step 1: The Administrator opens the Customers page.
Step 2: The Administrator enters a keyword in the search field.
Step 3: The system filters the customer records.
Step 4: The Administrator reviews the filtered results.
User Action: Administrator enters a customer name, email, or phone number in the search field.
System Response: The system displays the matching customer records.

Table 7
Use Case 7: Predictive Analytics
Use Case Goal: To allow the Administrator to view business forecasts and predictive insights.

Scenario 1: Administrator opens predictive analytics.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system.
Main Flow:
Step 1: The Administrator logs into the system.
Step 2: The Administrator opens the Predictive page.
Step 3: The system retrieves historical sales and inventory data.

Step 4: The system processes predictive analysis results.
User Action: Administrator clicks the Predictive menu.
System Response: The system displays forecast accuracy, predicted next month sales, restock items, and inventory turnover.

Scenario 2: Administrator views sales forecast.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system and forecast data must be available.
Main Flow:
Step 1: The Administrator opens the Predictive page.
Step 2: The Administrator reviews the sales forecast chart.
Step 3: The system displays projected sales values for future periods.
User Action: Administrator selects the forecast section.
System Response: The system shows the sales forecast for the next periods.

Table 8
Use Case 8: Manage Promotions
Use Case Goal: To allow the Administrator to create, monitor, and evaluate promotional campaigns.

Scenario 1: Administrator views promotion records.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system.
Main Flow:
Step 1: The Administrator logs into the system.
Step 2: The Administrator opens the Promotions page.
Step 3: The system retrieves the promotion records.
Step 4: The Administrator reviews the promotions list and analytics.
User Action: Administrator clicks the Promotions menu.

System Response: The system displays active promotions, revenue generated, effectiveness rate, and units affected.

Scenario 2: Administrator creates a promotion.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system and authorized to create promotions.
Main Flow:
Step 1: The Administrator opens the Promotions page.
Step 2: The Administrator clicks Create Promotion.
Step 3: The Administrator enters the promotion details.
Step 4: The Administrator saves the promotion record.
User Action: Administrator fills out the promotion form and clicks Save.
System Response: The system stores the promotion and updates the promotion list.

Scenario 3: Administrator views promotion performance.
Actor: Administrator
Pre-condition: The Administrator must be logged into the system and promotion data must exist.
Main Flow:
Step 1: The Administrator opens the Promotions page.
Step 2: The system retrieves campaign performance data.
Step 3: The Administrator reviews the charts and campaign records.
User Action: Administrator selects the promotion analytics section.
System Response: The system displays promotion performance comparison, category impact distribution, and discount effectiveness analysis.

Table 9
Use Case 9: Generate and Export Reports
Use Case Goal: To allow authorized users to generate and export reports from the system.

Scenario 1: Administrator generates reports.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and authorized to generate reports.

Main Flow:

Step 1: The Administrator logs into the system.

Step 2: The Administrator opens the Reports page.

Step 3: The system displays available report types and date filters.

Step 4: The Administrator selects the report period and report type.

Step 5: The system processes the request and generates the report.

User Action: Administrator selects the desired report options.

System Response: The system displays the generated report.

Scenario 2: Inventory Staff views reports and analytics.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and authorized to view reports.

Main Flow:

Step 1: The Inventory Staff logs into the system.

Step 2: The Inventory Staff opens the Reports & Analytics page.

Step 3: The system retrieves inventory and sales performance data.

Step 4: The Inventory Staff reviews the displayed reports.

User Action: Inventory Staff clicks the Reports & Analytics menu.

System Response: The system displays inventory turnover, units sold, average days to sell, monthly comparison, and top performing products.

Scenario 3: Administrator exports reports.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system and a report must already be generated.

Main Flow:

Step 1: The Administrator opens the Reports page.

Step 2: The Administrator reviews the generated report.

Step 3: The Administrator clicks Export Report.

Step 4: The system prepares the report file for download.

User Action: Administrator clicks the Export Report button.

System Response: The system downloads the generated report file.

Scenario 4: Inventory Staff exports reports.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system and a report must already be displayed.

Main Flow:

Step 1: The Inventory Staff opens the Reports & Analytics page.

Step 2: The Inventory Staff reviews the generated report.

Step 3: The Inventory Staff clicks Export Report.

Step 4: The system prepares the report file for download.

User Action: Inventory Staff clicks the Export Report button.

System Response: The system downloads the generated report file.

Table 10

Use Case 10: Log Out

Use Case Goal: To allow users to securely exit the system.

Scenario 1: Administrator logs out of the system.

Actor: Administrator

Pre-condition: The Administrator must be logged into the system.

Main Flow:

Step 1: The Administrator clicks the Logout button.

Step 2: The system ends the current session.

Step 3: The system redirects the Administrator to the login page.

User Action: Administrator clicks the Logout button.

System Response: The system logs out the Administrator and returns to the login page.

Scenario 2: Inventory Staff logs out of the system.

Actor: Inventory Staff

Pre-condition: The Inventory Staff must be logged into the system.

Main Flow:

Step 1: The Inventory Staff clicks the Logout button.

Step 2: The system ends the current session.

Step 3: The system redirects the Inventory Staff to the login page.

User Action: Inventory Staff clicks the Logout button.

System Response: The system logs out the Inventory Staff and returns to the login page.

Scenario 3: Sales Staff logs out of the system.

Actor: Sales Staff

Pre-condition: The Sales Staff must be logged into the system.

Main Flow:

Step 1: The Sales Staff clicks the Logout button.

Step 2: The system ends the current session.

Step 3: The system redirects the Sales Staff to the login page.

User Action: Sales Staff clicks the Logout button.

System Response: The system logs out the Sales Staff and returns to the login page.

3.9 Timetable (Gantt Chart)